Adaptive Learning Systems Using Deep Behavioural Analytics

A systematic, critical literature review and research-gap analysis

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How to read the evidence tags in this document

Every substantive claim carries a tag. The tags are not decoration — they mark exactly how far the claim was checked, so that nothing in this document is mistaken for a stronger result than it is.

- V **Verified.** The source page or abstract was retrieved directly during this review, and the claim comes from text actually read.
- S **Snippet-level.** Title, venue and URL were returned by an academic search, and the claim comes from an indexed snippet or an aggregator page. The full text was *not* independently verified.
- I **Interpretation.** My synthesis or inference across sources. Defensible, but it is argument, not a reported finding.
- H **Hypothesis.** A proposition for your research to test. It is not an established finding and must never be cited as one.

Three publishers (MDPI, ACM Digital Library, some Elsevier pages) returned HTTP 403 to automated retrieval during this review. Claims sourced from them are tagged S and explicitly marked "not independently verified". You must open those yourself before citing them. Citation counts were not retrievable for most items and have been omitted rather than estimated — an estimated citation count is a fabricated number.

Part 1 — Executive Summary

1.1 The current state of adaptive learning research

Adaptive learning research today sits on four largely separate pillars, and the separation between them is the single most important fact for your project.

Pillar one is psychometric. Item Response Theory and Computerized Adaptive Testing gave the field a rigorous, well-understood machinery for selecting the next item given an ability estimate. Critically, this tradition *already absorbed response time* two decades ago: van der Linden's hierarchical framework for modelling speed and accuracy jointly is a foundational, widely-extended model S, and joint response/response-time models are reported to improve the precision of ability estimation over accuracy-only models S. Response times have also been folded into item selection inside cognitive-diagnostic CAT S. This matters for your novelty claim: "response time improves the learner model" is not an open question in psychometrics. It is settled, and it was settled before deep learning arrived.

Pillar two is knowledge tracing. From Corbett and Anderson's Bayesian Knowledge Tracing S through Piech et al.'s Deep Knowledge Tracing S to attention-based models such as AKT S and SAINT+ V, the field optimises one target: predicting whether the next answer will be correct. Behavioural features entered this pillar too — SAINT+ adds elapsed time and lag time as embeddings and reports a 1.25% AUC gain over SAINT on EdNet V; LPKT builds its basic learning cell as the tuple *(exercise, answer time, answer)* with explicit learning and forgetting gates S. So "behavioural signals inside a deep knowledge tracing model" is also occupied territory.

Pillar three is behavioural and affective learning analytics. Ryan Baker's line of work built log-based detectors of gaming the system S, off-task behaviour S and wheel-spinning S; sensor-free affect detection infers boredom, frustration, confusion and engaged concentration purely from interaction logs, and is now mature enough to have its own systematic literature review V. Separately, psychometrics developed response-time-effort indices for detecting rapid guessing and disengaged responding S. This pillar is rich in *detection* and comparatively poor in *closed-loop action*.

Pillar four is sequential decision-making. Reinforcement learning and contextual bandits for instructional policy have a long history S and some genuinely deployed successes — a contextual-bandit curriculum evaluated in a randomised controlled trial reported superior completion rates and improved engagement V. But the RL-for-education literature is candid that simulators, sample efficiency and honest evaluation remain the binding constraints V (abstract-level only; the full-text discussion of these constraints was not retrieved).

1.2 How behavioural analytics is currently used

Three distinct usage patterns appear, and they are not equally represented.

(a) As auxiliary features inside a predictor. This is by far the most common. Response time, hint requests, attempt counts and lag time are appended to a knowledge tracing model to squeeze out AUC. A recent survey of deep-learning KT from a cognitive-processing angle states the position bluntly: behaviour-domain features that embed rich cognitive cues "are often treated merely as auxiliary attributes or as external factors influencing knowledge states", which limits the model's ability to characterise mastery from multiple perspectives S. That sentence is, in effect, an invitation written by the field to the exact project you are proposing.

(b) As the input to a standalone detector. Sensor-free affect detectors, off-task detectors, wheel-spinning detectors, disengagement detectors from log files, and MOOC dropout predictors from clickstream all live here. They typically stop at a validated classifier and a dashboard.

(c) As a driver of real-time adaptation. This is the thinnest layer. It exists — gaming detectors that trigger supplemental material reportedly halve gaming frequency S, and affect-adaptive tutoring systems that adjust difficulty from detected emotion are an active area S — but the affect-adaptive work overwhelmingly relies on *cameras and physiological sensors*, not on interaction telemetry alone.

1.3 Major research trends

1. **Saturation and a credibility correction in knowledge tracing.** The pyKT benchmark found that improper evaluation settings cause label leakage and performance inflation, and that "the improvement of many DLKT approaches is minimal compared to the very first DLKT model proposed by Piech et al." V. Gervet et al. found logistic regression with good features leads on moderate-size datasets, with DKT ahead only on large datasets or where precise temporal information matters, and Markov methods such as BKT lagging V. Automated feature search for logistic knowledge tracing has been reported to surpass deep learning in both accuracy and explainability S. **Any proposal of yours that promises "better AUC via a deeper model" walks straight into this literature and loses.**
2. **Movement from black-box to psychologically-grounded models.** Deep-IRT wraps DKVMN in an IRT layer for direct psychological interpretation while retaining DKVMN performance V; IKT uses three interpretable latent features with a Tree-Augmented Naive Bayes classifier and claims better prediction than deep student models with far fewer parameters V.
3. **Affect is being pulled inside the knowledge model.** DASKT extracts affective factors from non-affect-oriented behavioural data, clusters and models them spatiotemporally, and integrates simulated dynamic affect into knowledge state estimation, published in IEEE TKDE V. **This is the closest published relative of your proposal and you must read it first.**
4. **Multimodal learning analytics is expanding but is sensor-heavy.** The AI-in-MMLA systematic review covers 43 peer-reviewed papers from 2019–2024 S; the modalities in play are typically video, audio, gaze and physiology, with acknowledged problems of differing granularity and timescale alignment across sources S.
5. **Growing insistence on decision-relevance rather than prediction.** Recent work explicitly asks whether the interpretability of KT models actually supports teacher decision making, and calls for research into how pedagogical decisions can actually be improved from KT output S.

1.4 The strongest gaps, stated up front

Ranked, with full evidence in Part 5:

1. **The estimation–action gap.** Latent-state detectors and adaptive policies are developed in different papers, evaluated on different criteria, and almost never joined in one closed loop that is evaluated on learning outcomes rather than next-answer AUC.
2. **The unvalidated-latent-state gap.** Systems that claim to infer confidence, fatigue or cognitive load from clicks usually have no ground-truth labels for those constructs, making the claims unfalsifiable.
3. **The micro-interaction gap.** Mouse trajectory and keystroke dynamics are established uncertainty and stress signals in psychology and HCI, but are essentially absent from mainstream knowledge tracing and adaptive-sequencing pipelines.

The one-paragraph verdict on novelty

Your proposal as literally stated — "use behavioural signals to build a better learner model for adaptive learning" **— is not novel**. It has been done, repeatedly, with response time, attempts, hints and lag time, and it has been done inside transformer knowledge tracing, inside psychometric joint models, and inside affect-augmented KT. What is defensible is much narrower and is set out in Part 7. The honest classification of the contribution available to you is **novel feature combination + novel system architecture + evaluation contribution**, not novel algorithm. Full reasoning in "Would This Actually Be Novel?".

Part 2 — Taxonomy of Existing Research

Category A: Traditional Adaptive Learning — rule-based systems, IRT and CAT

How it works. Rule-based adaptive systems encode expert authored production rules: if mastery of prerequisite p is below threshold, present remediation. IRT models the probability of a correct response as a function of a latent ability θ and item parameters (difficulty, discrimination, guessing). CAT uses that model to select, at each step, the item that maximises information about the current θ estimate, terminating when the standard error is small enough. Cognitive-diagnostic CAT (CD-CAT) replaces the unidimensional θ with a discrete attribute-mastery profile, selecting items by criteria such as double Kullback–Leibler information or the GDI index [S].

Learner data used. Item responses; item parameters calibrated on a prior sample. Response time enters through the joint speed–accuracy tradition: van der Linden's hierarchical framework models responses and response times at a first level with a second level over person and item parameters, and permits plug-and-play choices of response and response-time distributions, estimated by MCMC [S]. Subsequent work reports that joint models improve the precision of ability estimation over accuracy-only models [S], and response times have been used inside CD-CAT item selection under a higher-order DINA model [S].

Adaptation decision. Deterministic and optimal with respect to an explicit information criterion. This is a real strength that machine-learning approaches often lack: the item choice is *derived*, not learned.

Strengths. Statistically principled; measurement error quantified; interpretable parameters; decades of operational deployment in high-stakes testing; response-time integration already solved.

Limitations. [I] Assumes stable ability within a session (van der Linden's model assumes constant speed and ability for the entire test [S]) — which is precisely the assumption that a fatigue-aware system must break. Item calibration is expensive. Learning during the test is not modelled: CAT measures, it does not teach. Rule-based systems, meanwhile, do not scale in authoring effort and encode a single designer's pedagogy.

> **Consequence for your project**
>
> [I] The intra-session non-stationarity of ability and speed is a genuine, literature-acknowledged weak point of the psychometric pillar. A fatigue/engagement-aware learner state is a coherent attack on it. But you must frame it that way — as relaxing a known modelling assumption — rather than as "adding new features".

Category B: Knowledge Tracing — BKT and its descendants

How it works. BKT is a two-state hidden Markov model per skill with four parameters: prior knowledge, learn rate, slip and guess; it treats student knowledge as a set of binary mastered/not-mastered variables [S]. Corbett and Anderson themselves attempted individualisation with mixed results, and the field made little progress on it until Pardos and Heffernan formulated individualisation entirely within a Bayesian-network framework, fitting individual and skill parameters simultaneously in one step [S]. Later work produced individualised BKT variants [S] and an accessible open implementation, pyBKT [S].

Learner data. Binary correctness sequences per skill, and in some variants hints and attempt counts. Behavioural signals are not native to BKT.

Adaptation. Mastery-threshold sequencing — the canonical rule advances a student once the posterior probability of mastery passes roughly 0.95. Almost every deployed intelligent tutoring system that uses BKT uses this rule.

Strengths. Interpretable parameters that map to cognitive theory; cheap; robust with small data; a direct, defensible adaptation rule attached to it.

Limitations. One skill at a time, so no transfer; binary knowledge; no forgetting in the base model; no behavioural signals; parameter identifiability problems. Gervet et al. found Markov process methods such as BKT lag behind other approaches on their nine-dataset comparison V.

Category C: Machine Learning and Deep Learning Approaches to Learner Modelling

DKT. Piech et al. applied an LSTM over the interaction sequence, maintaining a compressed numerical representation of latent knowledge, and reported roughly a 25% AUC gain over BKT S. That headline gain did not survive scrutiny: subsequent work showed the gain over BKT is not as substantial as reported (Xiong et al., 2016) and that classical methods given more flexibility can match or outperform DKT (Khajah et al., 2016; Wilson et al., 2016) S — *these three critique papers were identified through a secondary source and were not retrieved individually; treat the attributions as unverified until you check them.*

Attention and transformer models. SAKT applied self-attention directly to interactions S. AKT couples attention with interpretable components inspired by cognitive and psychometric models, using a monotonic attention mechanism and Rasch-model regularisation, reported to outperform prior methods by up to 6% AUC S. SAINT separates exercise and response streams in an encoder–decoder transformer. Notably, the AKT authors observed that SAKT did not outperform DKT in their experiments S.

Memory and interpretability variants. DKVMN introduced key–value memory; Deep-IRT applies an IRT layer on top of DKVMN to give a direct psychological interpretation of both students and items while retaining DKVMN's performance V. IKT goes further from deep learning entirely, using individual skill mastery, an ability profile and problem difficulty in a Tree-Augmented Naive Bayes classifier, claiming better prediction than deep student models without a huge parameter count V.

The credibility correction. This is the most important sub-story in Category C. pyKT found that wrong evaluation settings cause label leakage leading to performance inflation, that preprocessing in existing work is private and custom which prevents standardisation, and that improvement of many deep KT approaches over the original DKT is minimal V. Gervet et al. concluded that logistic regression with the right features leads on datasets of moderate size or with very many interactions per student, while DKT leads on large datasets or where precise temporal information matters V. Automated search over logistic knowledge tracing features has been reported to surpass deep learning in accuracy *and* explainability S.

> **Consequence for your project**
>
> I Two hard requirements follow. First, you must include a strong feature-engineered logistic baseline; omitting it is now a reviewer-fatal flaw, not a stylistic choice. Second, you must adopt a leakage-safe evaluation protocol — the pyKT protocol is the obvious reference — and say so explicitly in your methods.

Category D: Behavioural Learning Analytics

Detectors of unproductive behaviour. Baker and colleagues built detectors of gaming the system — systematic guessing and help abuse — across several tutoring systems, distinguishing types of gaming associated with different learning outcomes S; when gaming is detected, alerting the student and teacher plus supplemental material reportedly cut gaming frequency in half S. Off-task behaviour has been modelled from log files using time features, performance features and mouse movement features S. Wheel-spinning — extensive time and effort without mastery — was formalised for ASSISTments S and later extended to distinguish wheel-spinning from productive persistence in game-based learning S.

Response-time-based disengagement. Wise and Kong's response time effort (RTE) index classifies each response as rapid guessing or solution behaviour against a per-item time threshold, with five stated validity criteria including that RTE should correlate with other effort measures but *not* with academic ability, and that rapid-guess accuracy should be at chance [S]. A normative threshold treats RTE below 0.90 as meaningful disengagement [S]. Later work enhances response time thresholds with additional response behaviours to detect disengaged examinees [S].

Clickstream at scale. MOOC dropout and performance prediction from clickstream is a large sub-literature using fusion deep models over behaviour features [S], graph-based models over student actions [S], and interpretable multi-layer representation learning [S]. The recurring stated challenge is extracting meaningful behavioural representations from low-level clickstream and escaping subjective manual feature engineering [S].

Limitation of the whole category. [I] These are detectors, and detectors terminate in a dashboard or an alert. With the notable exception of the gaming-intervention result, the loop from detection to a changed instructional decision to a measured learning outcome is rarely closed in a single study.

Category E: Engagement, Confidence and Affect Detection

Sensor-free affect detection. The defining property is that affect is inferred exclusively from logs of interactions with computer-based learning environments — no cameras, no physiology. The systematic literature review by de Morais et al. identifies the most frequently detected affective states and reports consistent model performance, while listing as priorities: improving detector performance, accumulating more samples of underrepresented emotions, identifying additional emotions, creating a shared database of action logs and emotion labels, and making model source code publicly accessible [V] (abstract-level; the review does not state the paper count or accuracy figures in its abstract). Baker-lineage detectors for boredom, frustration, confusion and engaged concentration on ASSISTments have been used downstream to predict state test scores, college enrolment and STEM major choice [S] — a strong argument that these constructs are real and consequential, not artefacts.

The theoretical spine. D'Mello and Graesser's model of affect dynamics places cognitive disequilibrium at the centre: an impasse produces confusion, which if resolved catalyses deeper learning and if unresolved escalates to frustration and then boredom [S]. Confusion has been argued to be beneficial for learning under the right conditions [S], and the persistence of these states has been studied as a half-life [S]. **This distinction between productive and unproductive confusion is the single most important piece of theory for your adaptation policy**, because it means "detected difficulty" must not automatically trigger help — sometimes the correct action is to leave the learner in the impasse.

Confidence. The evidence here is more fragmented and this should temper your ambitions. In test-taking research, answer changes from wrong to right outnumber right to wrong (one study reports 55% wrong-to-right, 25% right-to-wrong, 20% wrong-to-wrong [S]), and changes are most likely to be successful when a student's *in-the-moment* confidence in the initial answer is low, while post-hoc reflective confidence has no predictive power [S]. Psychometrically, answer revisits have been modelled directly — the hierarchical speed–accuracy–revisits model [S] — and aberrant answer-change patterns are studied for test-security detection [S]. On the machine-learning side, one recent study removing self-reported confidence features from a behavioural-telemetry model reports a large performance drop (AUC-ROC falling by 0.229 to 0.649, accuracy down 16.1%), concluding that pre-execution behavioural telemetry alone has very limited independent predictive capability [S] — *this is a negative result directly relevant to your proposal and you should read it in full.*

Category F: Cognitive Load and Fatigue Detection

Physiological measurement dominates. Cognitive load estimation research is overwhelmingly EEG-based, with machine-learning classifiers over EEG features reaching 81.65–90.37% with a DNN versus 64.24–72.39% with an SVM in one study [S], LSTM models over EEG [S], and frameworks combining EEG, heart-rate variability and skin conductance [S]. Importantly, a recent study documents discrepancies between self-reported and EEG-based mental workload estimates [S] — which means "ground truth" for cognitive load is itself contested.

Interaction-based proxies exist but are thinner. Typing behaviour changes with task demand: when task demand increased, keystroke speed decreased along with other typing changes [S]; studies of mouse and keyboard dynamics during mental arithmetic report detectable cognitive stress [S]; keystroke dynamics have been evaluated as a real-world biomarker for mental fatigue [S]; and typing behaviour has been linked to whether a task is cognitively simple recall or more taxing analysis [S]. One study of 160 undergraduates reports significant correlations between direct instruction, external stimuli such as a visible timer, learners' cognitive and affective states, and their mouse and keystroke behaviours [S].

Fatigue as a modelled latent state. A nonlinear state-space model formalising fatigue, attention and learning as a continuous-time dynamical system — fatigue as a latent state under load–recovery dynamics, attention as a fatigue-coupled resource, learning as attention-mediated accumulation, with optimisation-based intervention adjusting load, pacing and recovery scheduling — was published in the MDPI journal *Computers* in 2025 [S]. **Not independently verified: the publisher returned HTTP 403 to automated retrieval. You must open this yourself.** The key question you need to answer from the full text is whether it was validated on real learners or only in simulation; the abstract-level material available did not establish this.

Disengagement over sessions. Log-file disengagement detection faces a timing problem stated crisply in that literature: using full sessions as the unit of analysis leaves no time to intervene, because disengagement could only be detected after around forty-five minutes, by which point most disengaged students had already logged out; splitting sessions into ten-minute sequences was the response [S]. [I] This is the exact latency constraint your fatigue module must be designed against, and it is a good, citable justification for per-item rather than per-session estimation.

Category G: Multimodal Learner Modelling

Definitions and fusion. Worsley's three fusion approaches — naïve/feature fusion, low-level fusion at very small timescales, and high-level fusion at a higher level of meaning — remain the standard vocabulary [S]. The community's own statement of challenges, from the LAK'17 data-challenges paper by Spikol, Prieto, Rodríguez-Triana, Worsley, Ochoa and Cukurova, includes differing granularity and the need to align timescales across sources [S]. A systematic review of multimodal data fusion in learning analytics exists in *Sensors* [S] — **not independently verified, publisher returned 403.** The AI-in-MMLA systematic review analysed 43 peer-reviewed papers from 11 databases published 2019–2024, finding a growing trend concentrated in tertiary education [S].

Modality bias. [I] The critical observation for your project is *which* modalities MMLA means. It overwhelmingly means video, audio, gaze, posture and physiology. Webcam eye tracking is being explored as the scalable alternative to research-grade trackers costing tens of thousands of dollars [S], but faces varying lighting and camera angles in unconstrained environments [S], limited resolution relative to research-grade equipment [S], and student unwillingness to be tracked by webcam on privacy and consent grounds [S]. Privacy-preserving computer-vision approaches for MMLA are an active LAK topic [S].

Multimodality without sensors. A small number of works treat KT itself as a multimodal fusion problem — for example combining multimodal fusion with neural architecture search applied to knowledge tracing [S]. [I] This framing, "multimodal within pure interaction telemetry", is under-occupied compared to sensor-based MMLA, and it is where your defensible territory lies.

Category H: Reinforcement Learning and Sequential Adaptation

Formulation. The tutoring system is the agent, the student is the environment, actions are instructional choices, and reward is learning gain — often with a penalty on the number of items assigned. Early work applied RL to pedagogical policy in adaptive and intelligent educational systems, with RLATES providing navigation support through content [S], and demonstrated that policies can be initialised from simulated students whose parameters capture decision time, help-request frequency and task accuracy [S].

Modern results. RL for the adaptive scheduling of educational activities was published at CHI 2020 S. An RL tutor was found to better support lower performers in a math task S — I a differential-effect result worth noting, since aggregate null effects can hide real subgroup effects. Contextual bandits are the practical middle ground: a model-based contextual bandit trained on trajectories from thousands of students and evaluated in a randomised controlled trial reported superior completion rates and significantly improved engagement V (specific completion-rate percentages were not stated in the abstract). Contextual bandit work also appears at LAK for student-support recommendation with expert features S, in an educational recommender using Linear Thompson Sampling reporting a 15.2% improvement in average skill gain over non-contextual baselines S, and in the MOOClet formalism which unifies experimentation, dynamic improvement and personalisation behind APIs supporting bandits and MDPs S.

Limitations. The survey by Singla, Rafferty, Radanovic and Heffernan frames the area's opportunities and challenges V — *only the abstract was retrieved; the specific discussion of simulator fidelity, sample efficiency and deployment constraints is in the full text and was not verified here.* I The structural problem is well known regardless: reward is delayed and noisy, exploration in education has an ethical cost because some students receive worse instruction, and policies learned against a simulator inherit every flaw of the simulator.

Part 3 — Detailed Literature Review Table

48 entries. Every title and URL below was returned by an academic search or retrieved directly during this review; none were composed from memory. Where a field is unknown it says "not stated in retrieved material" rather than being filled with a plausible guess. Evidence tag applies to the content of the row, not to the existence of the paper.

3.1 Psychometrics, IRT, CAT and response-time modelling

ID	Paper / link	Year	Research problem	Learner data	Behavioural features	Model	Adaptation method	Dataset	Key findings / limitations
P1	van der Linden, A Hierarchical Framework for Modeling Speed and Accuracy on Test Items, *Psychometrika* S	2007	Model responses and response times jointly	Item responses + response times	Response time (as a modelled outcome, not a feature)	Two-level hierarchical Bayesian; normal-ogive + lognormal RT; MCMC/Gibbs	Provides ability + speed estimates usable by CAT item selection	Test data (not stated in retrieved material)	Foundational; plug-and-play distributions. **Assumes constant speed and ability across the whole test** — the assumption a fatigue-aware model must break
P2	A Note on the Hierarchical Model for Responses and Response Times of van der Linden (2007), *Psychometrika* S	2014	Critique/refinement of P1	Responses + RT	Response time	Hierarchical IRT-RT	n/a (measurement)	n/a	Shows the joint model is actively contested and refined — cite when justifying model choice
P3	Joint Modeling of Ability and Differential Speed Using Responses and Response Times, *Multivariate Behavioral Research* S	2016	Allow speed to vary rather than be constant	Responses + RT	Within-test speed change	Joint IRT-RT	n/a	Not stated in retrieved material	Directly relevant: relaxes the constant-speed assumption. **Read this before claiming intra-session non-stationarity is unaddressed**
P4	Modeling Item Revisit Behavior: The Hierarchical Speed–Accuracy–Revisits Model S	2023	Model item revisits/answer revision psychometrically	Responses, RT, revisits	**Item revisits — i.e. answer changing**	Hierarchical extension of van der Linden	n/a (measurement)	Not stated in retrieved material	**Critical for your "option changes" feature: answer revision already has a formal psychometric model.** You cannot claim it as an unmodelled signal
P5	Utilizing response times in cognitive diagnostic CAT under the higher-order DINA model S	2019	Use RT in CD-CAT item selection	Responses + RT	Response time	Higher-order DINA + RT	**RT enters the item-selection rule directly**	Not stated in retrieved material	Precedent that behavioural data can drive real-time item choice, not just estimation
P6	Adapting cognitive diagnosis CAT item selection rules to traditional IRT, *PLOS ONE* S	2020	Transfer CD-CAT selection rules to IRT	Responses	None	DKL, GDI indices	Information-maximising selection	Simulation	Good baseline family for your "principled selection" comparison arm
P7	Wise & Kong, Response Time Effort — as reported in Enhancing response time	2005 / 2020	Detect non-effortful responding	Item RT	**Rapid guessing vs solution behaviour**	Threshold rule + RTE index	Motivation filtering (exclude	Large-scale assessment	Five validity criteria for effort indices — including RTE

ID	Paper / link	Year	Research problem	Learner data	Behavioural features	Model	Adaptation method	Dataset	Key findings / limitations
	thresholds with response behaviors for detecting disengaged examinees, *Large-scale Assessments in Education* S				**via per-item time threshold**		disengaged responses)		should *not* correlate with ability. RTE<0.90 = meaningful disengagement. **Use these criteria to validate your own engagement estimator**
P8	Wise, Rapid-Guessing Behavior: Its Identification, Interpretation, and Implications, *Educational Measurement: Issues and Practice* S	2017	Synthesise rapid-guessing methodology	Item RT	Rapid guessing	Threshold methods	n/a	Review	The standard reference for treating fast responses as disengagement rather than fluency
P9	Test engagement and rapid guessing: Evidence from a large-scale state assessment, *Frontiers in Education* S	2023	Prevalence and impact of rapid guessing at scale	Item RT	Rapid guessing	Threshold + IRT	n/a	State assessment	Recent, large-N evidence that engagement inferred from RT materially affects scores
P10	Response time as an indicator of test-taking effort in PISA: country and item-type differences S	2022	Does RT-as-effort generalise?	PISA RT	Response time	Threshold methods	n/a	PISA	**Effort–RT relationship varies by country and item type — a direct warning that RT thresholds are not universal constants**
P11	On the Optimality of the Detection of Examinees With Aberrant Answer Changes S	2018	Detect aberrant answer-change patterns	Answer-change logs	**Answer changes**	Statistical detection	n/a (test security)	Not stated in retrieved material	Shows answer-change data is already treated as a formal, analysable signal

3.2 Knowledge tracing: foundations, deep models and the credibility correction

ID	Paper / link	Year	Research problem	Learner data	Behavioural features	Model	Adaptation method	Dataset	Key findings / limitations
K1	Corbett & Anderson, Knowledge Tracing (as characterised in multiple retrieved sources; original not retrieved) S	1994/95	Model acquisition of procedural knowledge	Correctness per skill	None	2-state HMM per skill (prior, learn, slip, guess)	Mastery threshold (~0.95)	Cognitive Tutor	Foundational and still the default deployed policy. Binary knowledge, no transfer, no forgetting, no behaviour. **Original paper not retrieved — verify the citation yourself**
K2	Pardos & Heffernan, Modeling Individualization in	2010	Individualise BKT parameters	Correctness	None	Bayesian network, individual +	Mastery threshold	ASSISTments-lineage	Solved a problem open since Corbett & Anderson.

ID	Paper / link	Year	Research problem	Learner data	Behavioural features	Model	Adaptation method	Dataset	Key findings / limitations
	a Bayesian Networks Implementation of Knowledge Tracing, UMAP S					skill params fitted jointly			**Baseline for any "personalised parameters" claim you make**
K3	Individualized Bayesian Knowledge Tracing Models, AIED S	2013	Student-specific BKT	Correctness	None	Individualised BKT	Mastery threshold	Not stated in retrieved material	Authors not verified in retrieved material — check before citing
K4	pyBKT: An Accessible Python Library of Bayesian Knowledge Tracing Models S	2021	Reproducible BKT tooling	Correctness	Some variants use hints/attempts	BKT family	n/a	Multiple public	**Use this for your BKT baseline. Do not reimplement BKT**
K5	Piech et al., Deep Knowledge Tracing, NeurIPS S	2015	Sequence model of latent knowledge	Interaction sequence (skill, correctness)	None in base model	LSTM/RNN	Proposed exercise sequencing via expected knowledge gain	Khan Academy, ASSISTments, synthetic	Reported ~25% AUC gain over BKT; **gain later disputed.** Opaque hidden state; known consistency problems
K6	Xiong et al. 2016; Khajah et al. 2016; Wilson et al. 2016 — critiques of K5, *identified via secondary citation only, not retrieved* S	2016	Replicate/critique DKT	Correctness	None	BKT variants, IRT	n/a	Same benchmarks	Report that DKT's advantage over well-specified classical models is small or absent. **You must retrieve these individually — this row is the weakest-verified in the table**
K7	Ghosh, Heffernan & Lan, Context-Aware Attentive Knowledge Tracing, KDD S	2020	Attention + psychometric structure	Interaction sequence	None (uses item difficulty via Rasch)	Monotonic attention + Rasch regularisation	n/a (prediction only)	ASSISTments 2009/2015/2017	Up to 6% AUC over prior methods; authors note SAKT did not beat DKT in their runs. **Primary deep baseline for you**
K8	Shin et al., SAINT+: Integrating Temporal Features for EdNet Correctness Prediction, LAK V	2020/21	Add timing to transformer KT	Exercise + response streams	**Elapsed time and lag time, as embeddings fused into response embeddings**	Encoder–decoder Transformer	n/a (prediction only)	EdNet	**+1.25% AUC over SAINT.** This is the honest magnitude of a pure timing-feature gain in a strong model — anchor your expectations here, not at 25%
K9	Shen et al., Learning Process-	2021	Make knowledge state consistent	Interactions	**Answer time and interval**	Learning gate + forgetting	n/a (prediction	3 public datasets	Behaviour is architectural,

ID	Paper / link	Year	Research problem	Learner data	Behavioural features	Model	Adaptation method	Dataset	Key findings / limitations
	consistent Knowledge Tracing (LPKT), KDD S		with the learning process		**time as first-class parts of the learning cell**	gate over (exercise, answer time, answer)	+ more plausible knowledge state)		not auxiliary. **The closest architectural precedent to "behaviour-driven state estimation"**
K10	Yeung, Deep-IRT: Make Deep Learning Based Knowledge Tracing Explainable Using IRT V	2019	Explainability for DKVMN	Interactions	None	DKVMN + IRT output layer	n/a	Not stated in abstract	Retains DKVMN performance while giving psychological interpretation of student ability and item difficulty. **Template for making your state estimates explainable**
K11	Minn et al., Interpretable Knowledge Tracing: Simple and Efficient Student Modeling with Causal Relations, EAAI V	2022	Interpretable student model	Interactions	None	Tree-Augmented Naive Bayes over 3 latent features	n/a	Not detailed in abstract	Claims better prediction than deep student models with far fewer parameters. **Direct evidence that deep ≠ better**
K12	Liu et al., pyKT: A Python Library to Benchmark Deep Learning based Knowledge Tracing Models, NeurIPS D&B V	2022	Standardise KT evaluation	Interactions	Varies by model	Benchmark of many DLKT models	n/a	Many public datasets	**Wrong evaluation settings cause label leakage and performance inflation; improvement of many DLKT approaches over the original DKT is minimal.** Non-negotiable methodological citation for your paper
K13	Gervet, Koedinger, Schneider & Mitchell, When Is Deep Learning the Best Approach to Knowledge Tracing?, *JEDM* 12(3), 31–54 V	2020	When does deep learning actually win?	Interactions across 9 datasets	Features incl. temporal in the LR arm	LR vs DKT vs BKT, with ablations and calibration analysis	n/a	9 real-world datasets	**LR with the right features leads on moderate-size data; DKT leads on large data or where precise temporal information matters; BKT lags.** Defines exactly when your deep approach is justified
K14	Automated Search Improves Logistic Knowledge Tracing, Surpassing Deep	2024	Automated feature search for logistic KT	Interactions + engineered features	Feature set includes behavioural counts	Logistic KT + automated search	n/a	Public KT datasets	**A tuned logistic model beating deep KT on both**

ID	Paper / link	Year	Research problem	Learner data	Behavioural features	Model	Adaptation method	Dataset	Key findings / limitations
	Learning in Accuracy and Explainability, *JEDM* S								**accuracy and explainability. The hardest baseline you will face**
K15	Knowledge Tracing: A Survey, *ACM Computing Surveys* S	2023	Survey of KT	—	—	—	—	—	**Not independently verified: ACM DL returned 403.** Use as the canonical survey citation after you read it
K16	A Survey of Knowledge Tracing (arXiv preprint) S	2021	Survey of KT models and variants	—	—	—	—	—	**Preprint** — check whether the peer-reviewed version supersedes it before citing
K17	A survey of deep learning based knowledge tracing from a cognitive processing perspective, *Neurocomputing* S	2025	Recast KT around cognitive processing	—	Behaviour-domain features are the survey's central concern	—	—	—	**States that behaviour-domain features are "often treated merely as auxiliary attributes or as external factors", limiting multi-perspective mastery modelling. This sentence is the best single justification for your project's framing**
K18	Sun et al., DASKT: A Dynamic Affect Simulation Method for Knowledge Tracing, *IEEE TKDE*, DOI 10.1109/TKDE.2025.3526584 V	2025	Bring affect inside KT without affect labels	Interactions	**Affective factors extracted from non-affect-oriented behavioural data**	Clustering + spatiotemporal sequence modelling + KT	n/a (prediction; more plausible knowledge state)	2 public real-world datasets	**THE closest published relative of your proposal.** Simulates frustration, concentration, boredom, confusion dynamics and fuses them into knowledge state, beating SOTA KT. **Your novelty argument must explicitly differentiate from this paper**
K19	Enhanced Learning Behaviors and Ability Knowledge Tracing (LBAKT),	2025	Behaviour + ability in KT	Interactions	**Answer time, hint count, difficulty**	Attention-based KT	n/a	Public KT datasets	Ablation reported: removing attention+hint components

ID	Paper / link	Year	Research problem	Learner data	Behavioural features	Model	Adaptation method	Dataset	Key findings / limitations
	Applied Sciences S								dropped AUC to 0.77213 (−3.824%) and ACC to 0.72122 (−3.103%); removing difficulty cost only 0.43% AUC. **Not independently verified — MDPI returned 403**
K20	Precise modeling of learning process based on multiple behavioral features for knowledge tracing (MBFKT) S	2023	Multiple behavioural features in KT	Interactions	Multiple behavioural features	Multi-head attention + memory network + RNN	n/a	Public KT datasets	**Direct prior art for "multiple behavioural features fused in KT". Retrieve the published version and cite it, or a reviewer will**
K21	Deep learning knowledge tracing based on behavioral features S	2023	Behaviour features in DKVMN-family KT	Interactions	Response time, hints, attempts	DKVMN variant	n/a	ASSISTments 2009	Reported ~8% AUC improvement over the original DKVMN on ASSISTments 2009. **Venue and peer-review status not established from retrieved material — verify before citing**
K22	Advanced Knowledge Tracing: Incorporating Process Data and Curricula Information via an Attention-Based Framework, *JEDM* S	2024	Process data in KT with interpretability	Interactions + process data	**Process data**	Attention framework	n/a	Not stated in retrieved material	Peer-reviewed EDM-community precedent for process-data-driven KT
K23	Uncertainty-aware Knowledge Tracing (preprint) S	2025	Represent uncertainty in KT predictions	Interactions	Not established from retrieved material	Uncertainty-aware KT	n/a	Not stated	**Preprint.** Relevant if you want calibrated state estimates driving risk-sensitive actions
K24	Does Interpretability of Knowledge Tracing Models Support Teacher Decision Making? (preprint) S	2025	Do KT explanations change decisions?	KT output + teacher judgements	—	—	Human decision making	Teachers	**Preprint.** Calls for research into how pedagogical decision making can actually be improved from KT models — direct support for your evaluation gap

ID	Paper / link	Year	Research problem	Learner data	Behavioural features	Model	Adaptation method	Dataset	Key findings / limitations
K25	EdNet (dataset, via pyKT) S	2020	Large-scale KT benchmark	>131M interactions, >784k students, 2 years, from the Santa TOEIC ITS	**KT1 has timestamps; deeper subsets have richer behaviour**	—	—	EdNet	**Your primary public dataset for timing-based work. Note the domain is TOEIC prep in South Korea — generalisation is a real limitation to state**

3.3 Behavioural detection, affect, engagement, cognitive load and fatigue

ID	Paper / link	Year	Research problem	Learner data	Behavioural features	Model	Adaptation method	Dataset	Key findings / limitations
B1	Baker, Corbett, Roll & Koedinger, Detecting When Students Game the System S	2008	Generalisable gaming detector	Tutor logs	**Systematic guessing, help abuse, action timing**	Machine-learned detector	**Alerts + supplemental material on detection**	Cognitive Tutor	Distinguishes gaming types with different learning outcomes; **intervention reportedly halved gaming frequency — one of the few closed loops in this literature**
B2	Baker, Modeling and understanding students' off-task behavior in intelligent tutoring systems, CHI S	2007	Detect off-task behaviour	Tutor logs	**Time features, performance features, mouse movement features**	ML detector	n/a	Cognitive Tutor	**Early precedent for mouse features in ITS learner modelling — you cannot claim mouse data in ITS is new**
B3	Wheel-Spinning: Students Who Fail to Master a Skill, AIED S	2013	Formalise unproductive persistence	Practice sequences	Attempt counts over time without mastery	Detector	Implies stop/redirect	ASSISTments	The construct your "should we recommend prerequisite material" action needs
B4	Early Detection of Wheel-Spinning in ASSISTments S	2015	Detect wheel-spinning early enough to act	Practice logs	Attempts, correctness trajectory	Detector	Early intervention	ASSISTments	**Explicitly about detection latency — the same constraint your fatigue module faces**
B5	Detecting Wheel-spinning and Productive Persistence in Educational Games, EDM S	2019	Separate productive from unproductive persistence	Game logs	Persistence patterns	EDM detectors	n/a	Game-based learning	**Crucial nuance: persistence is not automatically bad. Your "struggling → give hint" rule must not punish productive persistence**
B6	de Morais et al.,	2023	State of sensor-free	Interaction logs only	Log-derived features	Advanced ML, consistent	Mostly none (detection	Multiple CBLEs	Stated priorities: better performance,

ID	Paper / link	Year	Research problem	Learner data	Behavioural features	Model	Adaptation method	Dataset	Key findings / limitations
	Automatic Sensor-free Affect Detection: A Systematic Literature Review (preprint) V		affect detection			performance reported	only)		more samples of underrepresented emotions, more emotions, **a shared database of action logs and emotion labels**, public code. **Preprint; paper count and accuracies not in the abstract**
B7	Sensor-Free or Sensor-Full: A Comparison of Data Modalities in Multi-Channel Affect Detection S	2016	Do sensors beat logs for affect?	Logs + sensor channels	Both	Affect detectors	n/a	ITS	**Read this to justify a sensor-free design: it is the direct comparison of exactly the tradeoff you are making**
B8	Improving Sensor-Free Affect Detection Using Deep Learning, AIED S	2017	Deep models for log-based affect	Interaction logs	Log features	Deep learning	n/a	ASSISTments-lineage	Precedent for deep sensor-free affect detection
B9	D'Mello & Graesser, Dynamics of affective states during complex learning, *Learning and Instruction* S	2012	Theory of affect dynamics in learning	Observations	—	Transition model	Implies affect-sensitive tutoring	Lab studies	**Cognitive disequilibrium → confusion → (resolved) deeper learning or (unresolved) frustration → boredom. The theoretical basis of your adaptation policy**
B10	Confusion can be beneficial for learning, *Learning and Instruction* S	2014	Is confusion good?	Experimental	—	—	Induced confusion via contradictions	Lab	**Directly contradicts the naive rule "detect difficulty → give help". Cite this when justifying a policy that sometimes withholds hints**
B11	The half-life of cognitive-affective states during complex learning, *Cognition & Emotion* S	2011	How long do affective states persist?	Observations	—	Survival analysis	n/a	Lab	**Gives you an empirical basis for the time constant of your latent-state smoothing rather than an arbitrary window**
B12	Cocea & Weibelzahl,	2009	Detect disengagement	Log files	Reading/answering times, session	ML classifiers	n/a	e-learning system	Foundational log-based disengagement

ID	Paper / link	Year	Research problem	Learner data	Behavioural features	Model	Adaptation method	Dataset	Key findings / limitations
	Log file analysis for disengagement detection in e-Learning environments S		from logs		structure				work
B13	Disengagement Detection in Online Learning Using Log File Analysis S	2016	Timely disengagement detection	Session logs	Session activity	ML	Enables earlier intervention	Online learning	**Full sessions gave detection only after ~45 minutes, by which time most disengaged students had logged out; ten-minute sequences fixed it. The clearest latency argument in this literature**
B14	Detecting Student Disengagement in Online Classes Using Deep Learning: A Review (preprint) V	2024	Review disengagement detection	Video, gaze, posture, mouse activity	Facial expression, eye movement, posture, **mouse activity as a non-face indicator**	Deep learning, computer vision, affective computing	Aimed at real-time monitoring	38 studies reviewed	**Preprint.** Confirms the field's camera-first bias and that non-face behavioural indicators are the minority approach
B15	Detection of Disengagement from Voluntary Quizzes: An Explainable ML Approach in Higher Distance Education (preprint) S	2025	Explainable disengagement detection	Quiz interaction logs	Irregular quiz behaviour, hesitation patterns	Explainable ML	n/a	Distance education	**Preprint.** Frames irregular quiz behaviour as hesitation/uncertainty — the nearest published treatment of your "hesitation" construct in a quiz setting
B16	Mouse tracking as a window into decision making, *Behavior Research Methods* S	2019	Methodology of mouse-tracking for cognition	Cursor trajectories	**Trajectory curvature, direction changes, attraction to non-chosen options**	Trajectory analysis	n/a	Lab tasks	**Harder decisions produce stronger curvature toward the non-chosen alternative. This is the mechanism your "hesitation from mouse" feature would rely on — and it was established in the lab, not the classroom**
B17	Discerning Mouse Trajectory Features With the Drift Diffusion Model,	2021	Link trajectories to evidence accumulation	Trajectories	Trajectory features	Drift diffusion model	n/a	Lab	**Gives you a principled generative account of hesitation rather than ad-hoc features. Strong candidate for a theory-grounded**

ID	Paper / link	Year	Research problem	Learner data	Behavioural features	Model	Adaptation method	Dataset	Key findings / limitations
	Cognitive Science S								**uncertainty estimator**
B18	Predicting respondent difficulty in web surveys: A machine-learning approach based on mouse movement features S	2020	Infer item difficulty from cursor behaviour	Mouse movement	**Mouse movement features**	Machine learning	n/a	Web survey respondents	**The closest methodological template for "infer perceived difficulty from mouse behaviour in a web form" — but in surveys, not learning**
B19	Using mouse cursor tracking to investigate online cognition: Preserving methodological ingenuity while moving toward reproducible science S	2021	Reproducibility of mouse-tracking	Cursor data	Many measures	—	n/a	Review	**A reproducibility warning: the proliferation of ad-hoc cursor measures is a known problem. Pre-register your feature definitions**
B20	Mouse Tracking Measures and Movement Patterns with Application for Online Surveys, CD-MAKE S	2018	Catalogue cursor measures	Cursor positions	**"Attraction" (area under curve), "zigzag" (direction changes)**	Descriptive + ML	n/a	>115,000 cursor positions from 300 users (as reported)	Concrete, implementable feature definitions for your instrumentation layer
B21	Monitoring Mouse Behavior in e-learning Activities to Diagnose Students' Acceptance S	2020	Mouse behaviour in e-learning	Mouse behaviour	Cursor patterns	—	n/a	e-learning	One of few mouse-behaviour studies actually set in e-learning rather than surveys or lab tasks
B22	The Effects of Typing Demand on Emotional Stress, Mouse and Keystroke Behaviours S	2015	Effect of demand on input behaviour	Keystrokes, mouse	**Inter-stroke time, keystroke speed, mouse behaviour**	Statistical analysis	Argues for adaptive e-learning use	160 undergraduates (as reported)	**Task demand and time pressure significantly affect mouse and keystroke behaviour; authors explicitly suggest use in adaptive e-learning. Strong support — and strong prior art**

ID	Paper / link	Year	Research problem	Learner data	Behavioural features	Model	Adaptation method	Dataset	Key findings / limitations
B23	Using Mouse and Keyboard Dynamics to Detect Cognitive Stress During Mental Arithmetic S	2015	Detect cognitive stress from input devices	Keyboard + mouse dynamics	Input dynamics	ML classification	n/a	Mental arithmetic task	**Cheaper and less invasive than physiological methods — the core practical argument for a sensor-free cognitive-load proxy**
B24	Detection of Mental Fatigue in the General Population: Feasibility Study of Keystroke Dynamics as a Real-world Biomarker, JMIR S	2024	Keystroke dynamics as a fatigue biomarker	Keystroke timing	**Keystroke dynamics**	ML	n/a	General population	**The strongest available evidence that fatigue is detectable from keystrokes alone — but outside education, and framed as a feasibility study**
B25	Utilizing linguistically enhanced keystroke dynamics to predict typist cognition and demographics, *IJHCS* S	2015	Predict cognition from typing	Keystrokes + linguistic context	Keystroke dynamics	ML	n/a	Typists	Typing behaviour distinguishes cognitively simple recall from taxing analysis. **Note the demographic-prediction angle — a fairness and privacy hazard for your system**
B26	Changing Answers in Multiple-Choice Exam Questions: Patterns of Top-Tier Versus Bottom-Tier Students S	2023	Who benefits from answer changes?	Exam answer changes	**Answer changes by direction**	Descriptive	n/a	Podiatric medical students	Reported 55% wrong→right, 25% right→wrong, 20% wrong→wrong. **Ability moderates the meaning of an answer change — your model must condition on ability, not treat changes uniformly**
B27	Should students change their answers on multiple choice questions?, *Advances in Physiology Education* S	2020/21	Confidence and answer revision	Answers + confidence ratings	Answer changes	Statistical	n/a	Students	**Changes succeed when in-the-moment confidence in the initial answer is low; post-hoc reflective confidence has no predictive power. Strong evidence that confidence must be captured in the moment — which is exactly what behavioural telemetry could do**

ID	Paper / link	Year	Research problem	Learner data	Behavioural features	Model	Adaptation method	Dataset	Key findings / limitations
B28	An Integrated ML and Hierarchical Variance Decomposition Pipeline for Student Performance Prediction and Metacognitive Calibration on Multi-Signal Telemetry (preprint) S	2026	Value of telemetry vs self-reported confidence	Behavioural telemetry + self-reports	Pre-execution telemetry	ML + variance decomposition	n/a	Not stated in retrieved material	**NEGATIVE RESULT, and the most dangerous paper for your thesis: removing self-reported confidence dropped AUC-ROC by 0.229 to 0.649 and accuracy by 16.1%, concluding pre-execution behavioural telemetry has very limited independent predictive capability. Preprint — verify, then address it head-on**
B29	A framework to estimate cognitive load using physiological data, *Personal and Ubiquitous Computing* S	2020	Cognitive load from physiology	Physiological signals	—	ML framework	n/a	Lab	Representative of the sensor-based mainstream you are deliberately not following
B30	Characterisation of Cognitive Load Using Machine Learning Classifiers of EEG Data, *Sensors* S	2023	Classify cognitive load from EEG	EEG	—	DNN vs SVM	n/a	Lab participants	DNN 81.65–90.37% vs SVM 64.24–72.39% accuracy (as reported). **Sets the accuracy bar sensor-free proxies are measured against — and shows why they will look weak**
B31	Discrepancies in Mental Workload Estimation: Self-Reported versus EEG-Based Measures (preprint) S	2025	Do subjective and neural load measures agree?	Self-report + EEG	—	Comparison	n/a	Visualization evaluation	**Preprint.** The two candidate ground truths for cognitive load disagree. **Cite this when you justify your choice of label — there is no uncontested gold standard**
B32	A Nonlinear State-Space Model for Fatigue Attention Dynamics in Online Learning	2025	Fatigue, attention, learning as one dynamical system	Not established from retrieved material	—	**Continuous-time nonlinear state-space; fatigue as latent state with load–recovery dynamics; attention as**	**Optimisation-based intervention: load regulation, pacing, recovery scheduling**	**Unknown — possibly simulation only**	**The single closest published relative for your fatigue module. NOT INDEPENDENTLY VERIFIED — MDPI returned 403. Your first task is to determine whether it was validated on**

ID	Paper / link	Year	Research problem	Learner data	Behavioural features	Model	Adaptation method	Dataset	Key findings / limitations
	Environments, *Computers* S					fatigue-coupled resource			real learners or only simulated ones; that answer decides whether a large part of your fatigue contribution survives

3.4 Multimodal learner modelling, sequential adaptation, effectiveness and ethics

ID	Paper / link	Year	Research problem	Learner data	Behavioural features	Model	Adaptation method	Dataset	Key findings / limitations
M1	Spikol, Prieto, Rodríguez-Triana, Worsley, Ochoa & Cukurova, Current and future multimodal learning analytics data challenges, LAK'17 S	2017	Community statement of MMLA challenges	Multimodal	—	—	—	—	**Differing granularity and timescale alignment across sources named as core challenges. Your fusion design must address both explicitly**
M2	Multimodal Data Fusion in Learning Analytics: A Systematic Review, *Sensors* S	2020	Review MMLA fusion	Multimodal	—	Fusion taxonomies	—	Systematic review	**Not independently verified — MDPI returned 403.** Likely your best single fusion-taxonomy citation once read
M3	A review on data fusion in multimodal learning analytics and educational data mining S	2022	Fusion in MMLA and EDM	Multimodal	—	Fusion methods	—	Review	Worsley's naïve / low-level / high-level fusion taxonomy is the standard vocabulary to position your design against
M4	Artificial intelligence in multimodal learning analytics: A systematic literature review, *Computers and Education: AI* S	2025	AI in MMLA, 2019–2024	Multimodal	—	Various	Some adaptive systems	43 papers, 11 databases	Growing trend, tertiary-education dominated; a portion of the reviewed work proposes adaptive systems from MMLA. **Your best "state of MMLA" citation**
M5	Exploring students' cognitive and affective states during problem solving through multimodal data, *JCAL* S	2022	Multimodal cognitive+affective states in programming	Multimodal	Interaction + sensor	Multimodal analysis	Analytic, not closed-loop	Programming activity	**Peer-reviewed precedent for jointly modelling cognitive and affective states — but as analysis, not as an adaptation driver**
M6	An Approach for Combining Multimodal Fusion and Neural Architecture Search Applied to	2021	Fusion + NAS for KT	Interactions, multiple modalities	Multiple	Fusion + NAS	n/a	KT datasets	**Preprint.** Direct prior art for "treat KT as a multimodal fusion problem" — your fusion novelty claim must be

ID	Paper / link	Year	Research problem	Learner data	Behavioural features	Model	Adaptation method	Dataset	Key findings / limitations
	Knowledge Tracing (preprint) S								narrower than this
M7	CVPE: A Computer Vision Approach for Scalable and Privacy-Preserving Socio-spatial, Multimodal Learning Analytics, LAK23 S	2023	Privacy-preserving MMLA	Video	—	Computer vision	n/a	Classroom	Shows the field treats privacy as a design constraint, not an afterthought
M8	Webcam-Based Eye Tracking to Detect Mind Wandering, *Behavior Research Methods* S	2023	Scalable gaze-based attention detection	Webcam gaze	Gaze	ML	n/a	Learners	**Read this to justify excluding webcam: it documents both the promise and the accuracy/privacy limits of the cheap alternative to eye trackers**
M9	Singla, Rafferty, Radanovic & Heffernan, Reinforcement Learning for Education: Opportunities and Challenges V	2021	Survey RL for education	—	—	RL family	Instructional policy	—	RL4ED workshop survey. **Only the abstract was retrieved; the detailed treatment of simulators, sample efficiency and deployment is in the full text and remains unverified here**
M10	Reinforcement Learning for the Adaptive Scheduling of Educational Activities, CHI S	2020	Schedule activities in real time	Activity outcomes	—	RL	**Assigns activity sequences maximising learning gain, minimising items assigned**	Online course	The reward structure you should copy: gain per item, not gain alone
M11	Iglesias et al., Learning teaching strategies in an Adaptive and Intelligent Educational System through Reinforcement Learning, *Applied Intelligence* S	2009	RL pedagogical policy (RLATES)	Navigation + performance	—	RL	Content navigation support	AIES	Foundational; policy improves purely from accumulated experience with similar learners
M12	Reinforcement learning tutor better supported lower performers in a math task, *Machine Learning* S	2023	Does an RL tutor help?	Task interactions	—	RL	Adaptive support	Math task	**Differential effect by prior ability. Design your analysis to detect subgroup effects — an aggregate null can hide a real benefit for low performers**
M13	Belfer, Kochmar & Serban, Raising Student Completion Rates with Adaptive Curriculum and	2022	Automated personalisation of learning experience	Trajectories from thousands of students	Not detailed in abstract	Model-based RL as contextual bandits	**Assigns learning activities**	Thousands of students; RCT	**Superior completion rates and significantly improved engagement in an RCT. The**

ID	Paper / link	Year	Research problem	Learner data	Behavioural features	Model	Adaptation method	Dataset	Key findings / limitations
	Contextual Bandits, AIED V								**gold-standard evaluation design for your adaptation claim (exact percentages not stated in the abstract)**
M14	Expert Features for a Student Support Recommendation Contextual Bandit Algorithm, LAK S	2024	Which context features for a bandit?	Student features	Expert-designed features	Contextual bandit	Support recommendation	Not stated in retrieved material	**Directly relevant: this is the question of whether your latent states are good bandit context**
M15	Lan & Baraniuk, A Contextual Bandits Framework for Personalized Learning Action Selection, EDM S	2016	Bandits for learning action selection	Learner state	—	Contextual bandit	**Learning action selection**	Not stated in retrieved material	The canonical formulation of "latent learner state as bandit context" — **your adaptation engine is a variant of this, so cite it rather than reinventing the framing**
M16	A Bandit-Based Approach to Educational Recommender Systems: Contextual Thompson Sampling for Learner Skill Gain Optimization, *INFORMS Transactions on Education* S	2025	Contextual bandit recommender	Learner context	—	Linear Thompson Sampling	Content recommendation	Not stated in retrieved material	Reported 15.2% improvement in average skill gain over non-contextual baselines. **Good precedent that context matters — and a number your work will be compared to**
M17	A Methodology for Discovering how to Adaptively Personalize to Users using Experimental Comparisons (MOOClet) S	2015	Unify experimentation and personalisation	Course interactions	—	Bandits, contextual bandits, MDPs behind APIs	Real-time modification of course components	Online courses	**The architectural pattern for running your adaptation as a continuous experiment rather than a fixed policy**
M18	Effectiveness of Intelligent Tutoring Systems: A Meta-Analytic Review S	2016	Do ITS work?	—	—	Meta-analysis	—	50 studies (as reported)	92% of studies favoured ITS; median effect sizes up to 0.66 (as reported). **Attributed to Kulik & Fletcher in secondary sources — verify authorship before citing**

ID	Paper / link	Year	Research problem	Learner data	Behavioural features	Model	Adaptation method	Dataset	Key findings / limitations
M19	A Meta-Analysis of the Effectiveness of Intelligent Tutoring Systems on College Students' Academic Learning S	2014	ITS effect on college learners	—	—	Meta-analysis	—	College students	Moderate positive effect, g ≈ .32–.37 (as reported). **The realistic effect size your system must be powered to detect**
M20	A Meta-Analysis of Moderators of the Effects of Technology-Enhanced Adaptive Learning on Primary and Secondary Students' Learning Outcomes, *JCAL* S	2026	What moderates adaptive-learning effects?	—	—	Meta-analysis	—	K-12	Recent; identifies moderators. **Use to choose your population and to argue why effects vary**
M21	Artificial intelligence in adaptive education: a systematic review of techniques for personalized learning, *Discover Education* S	2025	Review AI techniques for personalisation	—	—	Various	Various	Systematic review	Recent umbrella review for your related-work section
M22	Hakimi, Eynon & Murphy, The Ethics of Using Digital Trace Data in Education: A Thematic Review of the Research Landscape, *Review of Educational Research* S	2021	Ethics of digital trace data	Digital trace data	—	Thematic review	—	Review	**The single most relevant ethics citation for a system built entirely on digital traces. High-impact venue — do not omit it**
M23	Learning analytics and higher education: a proposed model for establishing informed consent mechanisms to promote student privacy and autonomy, *IJETHE* S	2019	Consent mechanisms for learning analytics	—	—	Conceptual model	—	—	**Gives you a concrete consent design to implement and cite, rather than a generic ethics paragraph**
M24	Beyond compliance: a Solove-informed analysis of tracking, profiling, and student privacy in learning management systems, *Frontiers in Education* S	2026	Privacy harms beyond consent	LMS behavioural data	—	Conceptual	—	—	**Argues privacy cannot be reduced to consent: surveillance, aggregation, secondary use, exclusion, distortion and decisional interference are distinct harms. Directly indicts fine-grained mouse/keystroke capture —**

ID	Paper / link	Year	Research problem	Learner data	Behavioural features	Model	Adaptation method	Dataset	Key findings / limitations
									address it, do not dodge it
M25	MOOC dropout prediction using a fusion deep model based on behaviour features, *Computers & Electrical Engineering* S	2022	Dropout from behaviour	Clickstream	**Behaviour features**	Fusion deep model	n/a	MOOC	Representative of behaviour-based prediction that stops at prediction
M26	Dropout Prediction over Weeks in MOOCs via Interpretable Multi-Layer Representation Learning (preprint) S	2020	Interpretable dropout prediction	Clickstream	Weekly behaviour	Multi-layer representation learning	n/a	MOOC	**Preprint.** Interpretability applied to behavioural representations — a pattern worth borrowing

Part 4 — Behavioural Signal Analysis

Signal-by-signal assessment. The "Reliability" column is the one that should govern your feature-selection decisions — several signals in your original list are far weaker than they look. IDs refer to Part 3.

Signal	Research	What it can indicate	How measured	Models used	Reliability / limitations	Use in your ALP
Response time (per item)	P1, P3, P5, P7–P10, K8, K9, K19 V / S	Ability + speed jointly; fluency; rapid-guessing disengagement; item difficulty	Timestamp difference from item render to answer submit	Hierarchical IRT-RT (MCMC); transformer embeddings; threshold rules	**Strongest and best-understood signal — and the most saturated.** Non-monotonic: fast can mean mastery *or* guessing, slow can mean effort *or* distraction. Thresholds vary by country and item type (P10). Needs per-item normalisation	**Include, but never raw.** Use log-RT standardised per item, plus an explicit rapid-guess flag using a per-item threshold. Do not present it as a novel feature
Time before first selection (decision latency)	B16, B17, B18 S	Deliberation vs impulsivity; retrieval difficulty	Render timestamp to first option click	Drift diffusion; ML on cursor features	Better-motivated than total RT because it isolates decision from reading and motor execution. **Confounded by reading time — a long stem inflates it**	**Include, and control for stem length/word count.** This is a cleaner uncertainty proxy than total RT
Time-on-task / session duration	B12, B13, B32 S	Engagement; cumulative fatigue exposure	Aggregated active time per session	ML classifiers; state-space (B32)	**Detection latency is the killer: session-level features only resolve after ~45 minutes, by which time disengaged students have already left (B13)**	**Use as a slowly-varying fatigue input at 5–10 minute resolution, never as a session-end feature.** Feeds the fatigue state only
Number of attempts	B3, B4, B5, K19, K21 S	Persistence; struggle; wheel-spinning vs productive persistence	Count of submissions per item	Detectors; KT feature	**Ambiguous by design: high attempts is the signature of both the most and least productive learners (B5).** Only interpretable jointly with correctness trajectory	**Include, but only as part of a wheel-spinning-style composite** (attempts × trend in correctness), not as a bare count
Option changes / answer changes	P4, P11, B26, B27 S	In-the-moment uncertainty; metacognitive revision; occasionally aberrance	Log every option selection event, not just the final answer	Hierarchical speed–accuracy–revisits (P4); descriptive statistics	**Already formally modelled in psychometrics (P4) — you cannot claim it as unexploited.** Meaning is moderated by ability (B26): the same change means different things for strong and weak students. Sparse — most items have zero changes	**Include, conditioned on ability.** Cite P4 and B27 explicitly. Expect sparsity to limit its contribution on any single item; it will matter in aggregate
Skipping	P7, P8, B12 S	Avoidance; perceived difficulty; disengagement	Item presented then navigated away without answer	Threshold + effort indices	Confounded with test-navigation strategy (skip-and-return is a taught exam technique, not disengagement)	**Include, but distinguish skip-and-return from skip-and-abandon** — otherwise you will label good strategy as disengagement
Mouse movement / cursor trajectory	B2, B16–B21 S	Decision uncertainty (curvature toward the non-chosen option); perceived difficulty; attention	Sample cursor positions at fixed rate; derive attraction (area under curve), zigzag (direction changes), pauses, velocity	Trajectory analysis; drift diffusion (B17); ML on movement features (B18)	**Established in the lab, thin in real classrooms.** Reproducibility concerns from proliferating ad-hoc measures (B19). **Absent entirely on touch devices and mobile — a coverage problem, not a noise problem.** Sampling at high rates produces large volumes	**The strongest candidate for genuine novelty in a learning context — treat it as your primary experimental signal, with pre-registered feature definitions from B20 and a theory-grounded variant from B17.** Be explicit that it only applies to desktop MCQ interaction

Signal	Research	What it can indicate	How measured	Models used	Reliability / limitations	Use in your ALP
Clickstream / navigation patterns	M25, M26, B12 S	Engagement; strategy; dropout risk	Event log of all UI interactions	Deep sequence models; graph models; representation learning	Rich but low-level; the field's own stated difficulty is extracting meaningful representations without subjective manual feature engineering	**Include as a learned sequence representation rather than hand-crafted counts**, since that is where the literature says the value is
Typing / keystroke dynamics	B22–B25 S	Cognitive load (keystroke speed falls as demand rises); stress; mental fatigue	Inter-stroke intervals, dwell/flight times, backspace rate	ML classification	**Only available when the learner types** — in an MCQ-dominated ALP this may be a small minority of items, creating severe missingness. Evidence base is mostly outside education (B23, B24). **Keystroke dynamics also predict demographics (B25) — a re-identification and fairness hazard**	**Include only if your platform has substantial free-text or numeric-entry items.** If it is MCQ-only, drop it and say so — a feature present on 5% of items cannot support a claim. Hash/aggregate rather than storing raw keystroke streams
Hint requests / help-seeking	B1, K19, K21 S	Help abuse vs appropriate help-seeking; struggle	Log hint events with timing	Gaming detectors; KT feature	Well-established. **Endogenous once your system adapts — if your policy changes hint availability, hint-request features become a function of your own policy**, which breaks naive offline evaluation	**Include, but treat as policy-dependent.** This is a concrete reason your offline evaluation needs importance weighting rather than plain replay
Performance trend across difficulty	K7, K13, K19 V / S	Ability, and the boundary of the zone of proximal development	Correctness conditioned on calibrated item difficulty	Rasch/IRT; attention with Rasch regularisation (K7)	Requires calibrated difficulty; cold-start problem for new items. Note LBAKT reports difficulty contributed only ~0.43% AUC in its ablation (K19)	**Essential — this is your knowledge state's backbone.** Use Rasch-style item parameters as in AKT
Inactivity / idle gaps	B2, B13 S	Off-task behaviour; disengagement; interruption	Gap between consecutive interaction events, with a configurable threshold	Off-task detectors	**Cannot distinguish "thinking hard" from "left the room" from a browser-tab switch.** This is a genuine, unresolved ambiguity	**Include with a visibility/focus signal from the browser to disambiguate** — that is a cheap engineering fix the literature often lacks
Lag time between sessions	K8, K9 V / S	Forgetting; spacing effects	Timestamp difference between consecutive interactions	SAINT+ embeddings; LPKT forgetting gate	Well-established and already exploited by the strongest KT models. **No novelty available here**	**Include as a baseline capability, not a contribution.** Omitting it would make your model weaker than SAINT+ for no reason

4.1 Which higher-level states are actually inferable, ranked by evidential support

Latent state	Support level	Basis	Recommendation
Knowledge / mastery	**Strong**	Thirty years of BKT, IRT and deep KT; well-defined observable proxy (future correctness)	**Keep. This is the anchor state; everything else is measured by whether it improves this one or improves decisions.**
Engagement / disengagement	**Strong**	RTE and rapid-guessing methodology with explicit validity criteria (P7–P10); log-based disengagement detection (B12–B15); Baker's gaming and off-task detectors (B1, B2)	**Keep. It has the best-defined validation protocol of any behavioural construct — use Wise & Kong's five criteria as your validation standard.**
Confusion (as an affective state)	**Moderate–strong**	Sensor-free affect detection is a mature literature with a systematic review (B6); grounded in D'Mello & Graesser's theory (B9–B11); DASKT already fuses it into KT (K18)	**Keep, but you are entering occupied territory. Your differentiation must be the action taken, not the detection.**
Confidence / uncertainty	**Moderate, and contested**	Answer-change research supports in-the-moment confidence as meaningful (B27); mouse trajectory curvature is a validated uncertainty index in the lab (B16, B17); **but B28 reports behavioural telemetry alone has**	**Keep only if you collect confidence labels.** Periodic in-the-moment confidence probes on a sample of items give you ground truth and directly answer B28. Without labels this construct is unfalsifiable.

Latent state	Support level	Basis	Recommendation
		very limited independent predictive power once self-report is removed	
Cognitive load	**Weak from interaction data alone**	Strong EEG evidence (B29–B31), weak and mostly non-educational interaction-based evidence (B22, B23); **and the two candidate ground truths disagree with each other (B31)**	**Demote.** Do not claim to estimate cognitive load. Either drop it, or reframe it honestly as "perceived item difficulty", which is what B18 actually demonstrates from mouse data and which you can label cheaply with a one-click difficulty rating.
Fatigue	**Weak-to-moderate, and thin in education**	Keystroke-based fatigue detection exists but is a feasibility study outside education (B24); B32 models fatigue as a latent state but its validation status is unverified	**Keep as a genuine gap — this is where your novelty is highest — but scope it to within-session performance and speed decay, which you can measure, rather than to fatigue as a psychological construct, which you cannot label.**
Hesitation	**Weak as a distinct construct**	Only appears in adjacent literature (B15 preprint, B16–B18); no standard operationalisation in education	**Do not model as a separate latent state. It is an observable feature (decision latency + trajectory curvature + option changes) that feeds confidence. Treating it as its own state adds parameters and no evidence.**

The reduction this analysis forces

I Your proposal lists six latent states. The literature supports **four**: knowledge, engagement, confidence (with labels) and a bounded within-session fatigue/decay state. Cognitive load should be reframed as perceived difficulty; hesitation should be demoted to a feature. **Cutting two states is not a weakening of the contribution — it is what makes the remaining four defensible**, because each surviving state has a validation route and a labelling strategy. A paper claiming six states with ground truth for none is a much weaker paper than one claiming four states with validation for each.

Part 5 — Research Gap Analysis

Eight gaps, each classified, each with an explicit novelty-risk rating. Three are strong, two are moderate, three are weak or already closed. **The weak ones are included deliberately: they are the claims you would most naturally make, and they would not survive review.**

Gap 1: The estimation–action gap (Real-time adaptation gap + Evaluation gap)

What existing research does. Latent-state estimation and instructional decision-making are pursued in separate literatures with separate success criteria. Knowledge tracing optimises next-answer AUC (K5–K21). Affect and behaviour detection optimises detector accuracy against observation labels (B1–B15). Sequential adaptation optimises completion or learning gain but typically conditions on a thin state — correctness history, or hand-designed context features (M10, M13–M17).

What is still missing. A system in which rich behaviourally-inferred states are the context for the action-selection policy, and in which the whole loop is evaluated on *learning outcomes and instructional efficiency* rather than on prediction accuracy. Almost no study shows that a better learner model produces better instructional decisions; the field assumes the implication rather than testing it.

Evidence from literature. pyKT shows that gains from many deep KT models over the original DKT are minimal, and that evaluation-protocol errors inflate reported performance [V] — so the accuracy currency itself is unreliable. Gervet et al. show that which model "wins" depends entirely on dataset properties [V]. A 2025 preprint asks directly whether KT interpretability supports teacher decision making and calls for research into how pedagogical decisions can actually be improved from KT output [S]. Meanwhile Baker's gaming intervention (detection → alert plus supplemental material → gaming frequency halved) [S] and the contextual-bandit RCT reporting improved completion and engagement [V] stand out precisely *because* they closed the loop — they are exceptions, not the norm.

Why this matters. If a richer learner state does not change what the system does, or changes it without improving outcomes, then behavioural analytics in education is an expensive measurement exercise. This is the question the field has been avoiding, and it is answerable.

How your ALP could address it. Hold the adaptation policy architecture fixed and vary only the state representation fed into it: correctness-only, correctness+timing, and full behavioural state. Measure post-test learning gain, items-to-mastery, and dropout. This is a clean, well-powered experiment that produces a publishable result *whichever way it comes out* — including a null result, which would itself be a valuable contribution given how much the field assumes.

Novelty risk: HIGH CONFIDENCE. The gap is real, is acknowledged in recent literature, and the experiment to close it is feasible at your scale.

Gap 2: Latent states asserted without ground truth (Methodological gap + Learner modelling gap)

What existing research does. The rigorous end of this literature does label its constructs. Sensor-free affect detection is trained against systematic field observations of boredom, confusion, frustration and engaged concentration V S. Rapid-guessing research validates its effort index against five explicit criteria, including that it should correlate with other effort measures and should *not* correlate with ability S. Cognitive load research validates against EEG or self-report S.

What is still missing. Applied adaptive-learning systems that claim to estimate confidence, fatigue or cognitive load from clicks routinely have *no* labels for those constructs. The "state" is a named intermediate layer of a network trained end-to-end on correctness. Calling that layer "confidence" is an interpretive act, not a measurement — and it is unfalsifiable.

Evidence from literature. The sensor-free affect review's stated priorities include the need for a shared database of action logs *and emotion labels* V — an explicit statement that labels are the bottleneck. DASKT's own framing acknowledges the problem it works around: affective evaluation in KT is under-explored "due to the non-affect-oriented nature of the data and budget constraints", and it therefore *simulates* affect rather than measuring it V. The mental-workload discrepancy study shows self-report and EEG disagree S, so even the label is contested. And B28 reports that removing self-reported confidence collapses model performance, concluding that pre-execution behavioural telemetry has very limited independent predictive capability S.

Why this matters. Without labels you cannot distinguish "our confidence estimator works" from "our extra parameters helped fit correctness". A reviewer will ask this in the first paragraph of their report, and there is no answer available after the fact.

How your ALP could address it. Because you are building the platform, you can instrument label collection cheaply and from the start — a one-click confidence rating on a random 10–15% of items, a periodic single-item fatigue/effort probe, an optional perceived-difficulty rating, and if feasible a small human-observation study on a subsample. That converts every latent state from an assertion into a measurable quantity with a reported validation metric.

Novelty risk: HIGH CONFIDENCE as a methodological contribution. *Note that the contribution here is rigour, not architecture* — and rigour is a legitimate and increasingly valued contribution in this field, as the pyKT paper demonstrates.

Gap 3: Micro-interaction signals are absent from adaptive learning pipelines (Data gap + Multimodal fusion gap)

What existing research does. Mouse trajectory analysis is a validated window into decision uncertainty — harder decisions produce greater curvature toward the non-chosen alternative [S] — with a principled drift-diffusion account [S] and a demonstrated ML application predicting respondent difficulty from mouse movement in web surveys [S]. Keystroke dynamics track task demand and mental fatigue [S]. Baker used mouse movement features in off-task detection as early as 2007 [S].

What is still missing. These signals sit in HCI, psychology and survey methodology. They are essentially absent from knowledge tracing and from adaptive item-sequencing. The behavioural features in modern KT are almost entirely *timing and counting* features: elapsed time, lag time, attempts, hints (K8, K9, K19–K21). Nobody has shown whether within-item motor behaviour adds anything once elapsed time is already in the model.

Why this matters. This is the sharpest empirical question in your whole project, and it has a clean answer: *does trajectory curvature carry information about the learner's state beyond what response time already carries?* Response time is a scalar summary of a process; the trajectory is the process. If the answer is yes, that is a genuine finding. If the answer is no, that is also a genuine finding and it usefully closes a line of speculation.

How your ALP could address it. Instrument cursor sampling with pre-registered feature definitions taken from the existing catalogue (attraction/area-under-curve, zigzag/direction changes, pauses, velocity profile) [S], then run a strict incremental-value ablation against a model that already contains response time, decision latency, attempts and answer changes.

Novelty risk: HIGH CONFIDENCE for the question; MODERATE for the outcome. The gap in the literature is real and I found nothing occupying it. But be clear-eyed: **the result may well be that curvature adds little beyond response time**, and the desktop-only limitation is a genuine scope constraint you must state, not bury.

Gap 4: Fatigue and within-session non-stationarity are barely modelled in adaptive sequencing (Learner modelling gap)

What existing research does. Psychometrics assumes constant speed and ability across a test in the foundational joint model [S], with later work relaxing this to differential speed [S]. Disengagement detection over sessions exists but resolves too slowly to act on [S]. A 2025 state-space model treats fatigue as a latent state with load–recovery dynamics driving pacing and recovery scheduling [S] — **validation status unverified**. Keystroke-based fatigue detection exists outside education [S].

What is still missing. Knowledge tracing models — including the behaviour-augmented ones — model *forgetting between* sessions (via lag time) but not *degradation within* a session. A learner in minute 40 is treated identically to the same learner in minute 5. And no widely-cited adaptive system takes "recommend a break" as an available action with measured consequences.

Why this matters. Within-session decay is measurable without any new construct: response accuracy and speed on items of matched difficulty as a function of time-in-session. It is a real modelling omission with a concrete observable.

How your ALP could address it. Add a within-session state driven by time-on-task, decay in matched-difficulty accuracy, and speed drift; allow "suggest a break" and "reduce difficulty" as policy actions; evaluate whether break recommendations improve post-break performance and session-level efficiency.

Novelty risk: MODERATE CONFIDENCE. Conditional on B32. If that state-space paper turns out to be validated on real learners with an intervention study, this gap shrinks substantially and you must reposition to "empirical validation in a deployed ALP" rather than "first model of". **Read it before you write your proposal.**

Gap 5: No unified multi-state learner model driving action selection (System architecture gap)

What existing research does. Individual states are modelled well in isolation. DASKT is the closest integration, fusing simulated affect dynamics into knowledge state within one model V. Multimodal learning analytics studies cognitive and affective states together but analytically rather than as an adaptation driver S. Contextual bandit work formalises "learner state as policy context" S.

What is still missing. A single architecture that jointly estimates knowledge, engagement, confidence and within-session fatigue from one interaction stream and passes all of them as context to an action-selection policy with an explicit, inspectable mapping from state to action.

Why this matters. The states interact in ways that change the correct action. Low confidence with high engagement warrants a hint. Low confidence with low engagement warrants a difficulty reduction or a break, because a hint to a disengaged learner is help-abuse bait S. Confusion with rising accuracy is productive and should be left alone S. **These interaction rules are exactly what a single-state system cannot express, and they are the real argument for a multi-state model.**

How your ALP could address it. A shared behavioural encoder with separate calibrated state heads, feeding a constrained policy whose state→action mapping is auditable.

Novelty risk: MODERATE CONFIDENCE. Integration is a legitimate systems contribution, but "we combined known components" is a weak claim unless the *interaction effects* are the finding. **Make the interaction rules the contribution, not the block diagram.** A block diagram is not a research result.

Gap 6: Explainability of behaviour-driven adaptation decisions (Explainability gap)

What existing research does. Explainability work in this area targets the *prediction*: Deep-IRT gives psychological interpretation of ability and difficulty V; IKT uses three interpretable latent features V; automated logistic-KT search claims deep-learning-beating accuracy *and* explainability S.

What is still missing. Explanations of the *decision*: "you are seeing an easier item because your accuracy on matched-difficulty items fell 18% over the last 20 minutes while your response times rose." A 2025 preprint asks whether KT interpretability actually supports teacher decisions and finds the question under-researched S.

Why this matters. A behaviour-tracking system that silently reroutes a learner based on inferred fatigue is precisely the surveillance-and-decisional-interference pattern the privacy literature warns about S. Explanation is the mitigation, not a feature.

How your ALP could address it. Because your states are separately estimated and separately labelled (Gap 2), each decision decomposes into named, validated contributions. Explanation becomes structurally available rather than retrofitted.

Novelty risk: MODERATE CONFIDENCE as a secondary contribution, LOW as a primary one. Explainable education AI is crowded. This strengthens your paper; it cannot carry it.

Gap 7: "Behavioural features improve knowledge tracing" (CLOSED — do not claim this)

What existing research does. SAINT+ adds elapsed and lag time as embeddings, gaining 1.25% AUC on EdNet V. LPKT makes answer time and interval time architectural, with learning and forgetting gates S. LBAKT reports ablations where removing attention and hint components costs 3.8% AUC S. MBFKT fuses multiple behavioural features with multi-head attention and memory networks S. A DKVMN variant with behavioural features reports ~8% AUC gain on ASSISTments 2009 S. A 2025 Neurocomputing survey is organised around exactly this theme S.

What is still missing. Nothing that would support a novelty claim. **This is your proposal's stated core, and it is done.**

Why this matters. If your paper's central claim is "we added behavioural features to a knowledge tracing model and AUC improved", it will be desk-rejected or dismantled in review, with SAINT+, LPKT and MBFKT cited against you. Worse, the honest expected gain is around 1–4% AUC, which is inside the range pyKT attributes to protocol differences and leakage V.

How your ALP could address it. It should not try. Use this as a *replication* component: reproduce the known effect under a leakage-safe protocol as the foundation on which the real contributions (Gaps 1–3) rest.

Novelty risk: NO NOVELTY. Explicitly closed.

Gap 8: Privacy-proportionate behavioural sensing (Privacy / ethical gap)

What existing research does. Privacy scholarship in learning analytics is well developed: digital-trace ethics has a *Review of Educational Research* thematic review S; consent-mechanism models exist S; and recent work argues privacy cannot be reduced to consent, naming surveillance, aggregation, secondary use, exclusion, distortion and decisional interference as distinct harms S. Privacy-preserving MMLA is an active LAK topic S.

What is still missing. An empirical account of the *tradeoff*: how much adaptation quality is lost by progressively removing invasive signals. Nobody has published a curve from "correctness only" through "plus timing" to "plus full motor telemetry" measured in learning outcomes.

Why this matters. Keystroke dynamics predict demographics S, making them quasi-biometric. Without a demonstrated benefit, collecting them is not justifiable. **And the ablation you need for Gap 1 and Gap 3 produces this curve for free.**

How your ALP could address it. Report your feature ablation as a privacy–utility curve. Same experiment, second contribution.

Novelty risk: MODERATE CONFIDENCE. Framing is novel and cheap for you to produce; it will not carry a paper alone but it is a strong differentiator and it pre-empts the ethics objection rather than absorbing it.

5.1 Gap summary

#	Gap	Category	Novelty confidence	Verdict
1	Estimation–action gap	Real-time adaptation + Evaluation	**High**	**Make this the thesis.** Publishable either way
2	Unvalidated latent states	Methodological + Learner modelling	**High**	**Make this the method.** Requires label collection from day one
3	Micro-interaction signals absent from adaptive pipelines	Data + Multimodal fusion	**High** (question) / Moderate (outcome)	**Make this the experiment.** Desktop-only scope must be stated
4	Within-session fatigue / non-stationarity	Learner modelling	Moderate — conditional on B32	Strong secondary; verify B32 first

#	Gap	Category	Novelty confidence	Verdict
5	Unified multi-state model driving actions	System architecture	Moderate	Contribution is the state-interaction rules, not the diagram
6	Explainable adaptation decisions	Explainability	Moderate secondary / Low primary	Supporting contribution only
7	Behavioural features improve KT	—	**None — closed**	**Do not claim. Replicate instead**
8	Privacy–utility curve for behavioural sensing	Privacy / ethical	Moderate	Free byproduct of the Gap 1/3 ablation

Part 6 — Direct Comparison With Your Proposed System

"Existing adaptive systems" and "behavioural analytics systems" are named specifically, per your instruction not to generalise. IDs refer to Part 3. The final column states what your system would need to do *differently*, not merely that it would have the capability.

Capability	Existing adaptive systems (KT / IRT-CAT / RL-bandit)	Behavioural analytics systems (detectors / MMLA)	Your proposed ALP — required differentiation
Correctness tracking	**Yes, universally.** BKT (K1–K4), DKT (K5), AKT (K7), IRT/CAT (P1, P6)	Yes, as a feature in detectors (B1–B5)	Yes — **no differentiation available. Baseline capability**
Response time	**Yes.** Psychometrics since van der Linden 2007 (P1, P3, P5); SAINT+ elapsed/lag time (K8); LPKT answer/interval time (K9); LBAKT (K19)	**Yes, centrally.** RTE and rapid-guessing (P7–P10); log-based disengagement (B12, B13)	Yes — **no differentiation. Any claim of novelty here is false**
Attempts	Yes, as a KT feature (K19–K21)	**Yes, centrally.** Wheel-spinning (B3–B5); gaming detection (B1)	Yes — differentiation only if used as a **productive-vs-unproductive persistence composite** rather than a count
Option changes	**Yes, in psychometrics.** Hierarchical speed–accuracy–revisits model (P4); aberrant answer-change detection (P11). **Rare in KT**	Studied descriptively (B26, B27) but not as a live system input	**Partial differentiation:** using option-change events as a real-time input to an *adaptation* decision, conditioned on ability. Must cite P4 and not claim the signal is unmodelled
Mouse behaviour	**Essentially no.** One early exception: off-task detection using mouse features (B2, 2007)	**Yes, but outside education.** Decision uncertainty (B16, B17); survey difficulty prediction (B18); e-learning acceptance (B21)	**Genuine differentiation. Strongest single novelty claim available** — within-item trajectory features as input to knowledge/confidence estimation and item selection. Desktop-only scope must be declared
Typing behaviour	**No**	**Yes, outside education.** Typing demand and stress (B22, B23); mental fatigue biomarker (B24); cognition prediction (B25)	**Differentiation available but conditional on item format.** If your ALP is MCQ-dominant, drop this and say so — a feature present on a small minority of items cannot support a claim
Confidence estimation	**No** in mainstream KT. Uncertainty-aware KT is a 2025 preprint (K23)	**Partial and contested.** Answer-change/confidence research (B26, B27); B28 reports telemetry alone has very limited independent power without self-report	**Differentiation only if you collect in-the-moment confidence labels.** Without labels this is an unvalidated claim, and B28 is direct evidence against it
Engagement estimation	**Rarely inside the model.** Bandit work uses engagement as an *outcome*, not a state (M13)	**Yes, maturely.** RTE (P7–P10); log-file disengagement (B12–B15); off-task and gaming detection (B1, B2)	Yes — **differentiation is using engagement as policy context rather than as a dashboard metric or an evaluation outcome**
Fatigue estimation	**No.** Foundational psychometric models assume constant speed and ability across a test (P1)	**Thin.** Keystroke fatigue biomarker outside education (B24); nonlinear state-space fatigue model (B32, **validation unverified**)	**Genuine differentiation if scoped to within-session decay in matched-difficulty accuracy and speed.** Conditional on what B32 turns out to have done
Cognitive load estimation	**No**	**Yes, but sensor-based.** EEG classifiers (B29–B31); interaction-based proxies are weak and non-educational (B22, B23)	**Recommend dropping the claim.** Reframe as perceived item difficulty (B18-style), which is labellable with one click and defensible
Multimodal behavioural fusion	**Partial.** Fusion + NAS applied to KT (M6, preprint); multiple-behavioural-feature KT (K20, K21)	**Yes, but sensor-modal.** MMLA means video, audio, gaze, physiology (M1–M5, M8)	**Differentiation is the modality set, not the act of fusing:** "multimodal within pure interaction telemetry, no cameras or physiology". Under-occupied and privacy-defensible
Real-time learner state estimation	**Yes for knowledge.** All KT models estimate state per interaction	**Yes for single constructs**, though session-level detectors resolve too slowly to act on (B13)	**Differentiation is multi-state and per-item latency**, explicitly designed against the ~45-minute detection-latency problem B13 documents

Capability	Existing adaptive systems (KT / IRT-CAT / RL-bandit)	Behavioural analytics systems (detectors / MMLA)	Your proposed ALP — required differentiation
Real-time content adaptation	**Yes.** Mastery-threshold sequencing (K1); CAT item selection (P5, P6); RL scheduling (M10); contextual bandits in an RCT (M13, M15–M17)	**Rare.** The main exception is gaming detection triggering supplemental material (B1)	**No differentiation in the act of adapting.** The differentiation is exclusively in *what state the adaptation is conditioned on* — this is Gap 1 and it is the whole thesis
Explainable adaptation	**Explainable prediction, not explainable decisions.** Deep-IRT (K10); IKT (K11); logistic KT (K14); K24 asks whether it helps decisions at all	**Partial.** Explainable disengagement detection (B15, preprint); interpretable dropout prediction (M26)	**Differentiation: decision-level explanation decomposed over separately-validated states.** Secondary contribution — the area is crowded
Evaluated on learning outcomes (added row — the decisive one)	**Mostly no.** KT evaluates AUC/ACC. Exceptions: bandit RCT (M13), RL tutor (M12), ITS meta-analyses (M18–M20)	**Almost never.** Detectors evaluate detector accuracy. Exception: gaming intervention (B1)	**This is where your paper is won.** Evaluate the loop on learning gain and items-to-mastery, not AUC

Part 7 — Recommended Research Direction

7.1 What the exact research problem should be

> Adaptive learning systems select the next instructional action from a learner model built almost entirely from answer correctness and coarse timing. Rich behavioural signals available in the same interaction stream are used for post-hoc analytics or for marginal gains in next-answer prediction, but it has not been established that they improve the *instructional decisions* a system makes, nor that the latent learner states inferred from them are valid measurements rather than convenient labels for network activations.

`I` Note what this framing avoids. It does not claim behavioural features are new. It does not claim a new architecture. It targets the untested inference — model quality implies decision quality — that the whole field rests on.

7.2 Primary research question

> **Does conditioning adaptive instructional decisions on a validated multi-state behavioural learner model — knowledge, engagement, confidence and within-session fatigue — produce measurably better learning outcomes and instructional efficiency than conditioning on correctness and response time alone?**

7.3 Sub-research questions

1. **Validity.** To what extent can engagement, confidence and within-session fatigue be estimated from interaction telemetry alone, when validated against independently collected labels (in-the-moment confidence probes, effort probes, rapid-guessing indices)? `I` *This must come first. Every later question presupposes it.*
2. **Incremental information.** Does within-item motor behaviour — cursor trajectory features, decision latency, option-change sequences — carry information about learner state beyond what response time, attempts and correctness already provide? `I` *This is the sharpest empirical question you have and it is answerable offline before any deployment.*
3. **Decision impact.** Holding the policy architecture fixed, does enriching the state representation change which actions are selected, and does it improve learning gain and items-to-mastery?
4. **Privacy–utility tradeoff.** How much adaptation quality is retained as progressively more invasive signal classes are removed?

7.4 The strongest gap

Gap 1, the estimation–action gap, made rigorous by Gap 2 (validated states) and made empirically sharp by Gap 3 (micro-interaction signals). `I` Gap 1 alone risks being read as an engineering evaluation; Gap 3 alone risks being a feature-engineering paper. Together they are a coherent thesis with a methodological backbone.

7.5 Which behavioural signals to include

Include — core	Include — experimental	Include only if format allows	Exclude or reframe
Correctness; item difficulty (Rasch-calibrated); log response time standardised per item; decision latency to first selection; attempts; lag time between sessions; rapid-guess flag; idle gaps with browser focus/visibility state	Cursor trajectory features (attraction/area-under-curve, zigzag/direction changes, pause count, velocity profile); option-change sequences conditioned on ability; within-session matched-difficulty accuracy and speed decay	Keystroke dynamics — **only if a substantial share of items require typed or numeric input.** Otherwise exclude and state why	**Cognitive load** → reframe as perceived item difficulty with a one-click label. **Hesitation** → demote to a feature feeding confidence, not a separate state. **Webcam, gaze, physiology** → exclude by design; cite the cost, accuracy and consent evidence (M8) as justification

7.6 Which signals are too speculative to validate

- **Cognitive load from clicks.** Its own literature cannot agree on a ground truth (B31), and the interaction-based evidence is non-educational (B22, B23). You would be estimating a construct with no defensible label.
- **Confidence without labels.** B28 is a direct negative result. I Either collect probes or drop the construct — there is no third option that survives review.
- **Emotion categories (frustration, boredom) without observation labels.** The sensor-free affect literature is built on systematic field observation. Claiming these states from an unlabelled network layer invites direct comparison with a mature literature you have not matched.
- **Keystroke dynamics in an MCQ platform.** Not speculative in principle — B24 is real — but unusable in practice at low coverage, and it carries demographic-inference risk (B25).

7.7 Machine learning, deep learning, knowledge tracing, or hybrid?

Hybrid, and specifically: a shared sequence encoder with separately-supervised, calibrated state heads feeding a contextual bandit policy.

- **Not pure deep KT.** pyKT and Gervet et al. show the marginal returns are small and dataset-dependent, and that a well-tuned logistic model is a serious competitor V S.
- **Not pure classical.** You need a sequence model to handle raw trajectory and clickstream representations, which is exactly the regime Gervet et al. identify as favouring deep models — large data and precise temporal information V.
- **Separately supervised heads, not one end-to-end objective.** This is the architectural consequence of Gap 2: a head trained against confidence probes is a measurement; a head that merely exists is not.
- **Contextual bandit, not full RL.** Bandits have deployed evidence in education (M13, M15–M17), need far less data than full RL, and support offline policy evaluation. I Full RL requires a student simulator, and a policy learned against a flawed simulator inherits its flaws — an avoidable risk at your scale.
- **Follow Deep-IRT's pattern** (K10) of an interpretable output layer over a neural encoder, so state estimates stay psychologically meaningful.

7.8 Baselines you must compare against

Baseline	Why it is required	Consequence of omitting it
BKT via pyBKT (K4)	The deployed-in-practice standard	Reviewers ask whether you beat what schools actually run
Feature-engineered logistic KT / PFA (K13, K14)	Gervet et al. and the automated-search JEDM paper show it can beat deep KT	**Fatal.** This is now the single most common reason KT papers are rejected
DKT (K5)	The reference point pyKT uses to show most gains are minimal	No anchor for interpreting your gain
AKT (K7)	Strong attention baseline with psychometric grounding	Your deep model looks untested against the real state of the art
SAINT+ or LPKT (K8, K9)	**These already use behavioural timing.** They are the true competitors for your central claim	**Fatal for the novelty argument.** Without these, "behaviour helps" is compared only against models that never had behaviour
Mastery-threshold policy (K1)	The deployed adaptation policy, and a strong one	Your policy gain is uninterpretable
IRT/CAT maximum-information selection (P6)	The principled psychometric selection rule	Reviewers from the measurement community dismiss the work
Random and fixed-order sequencing	Floor conditions	No evidence adaptation beats no adaptation at all
Ablation ladder: correctness-only → +RT → +attempts/changes → +trajectory	Isolates the contribution of each signal class	**Without this, you cannot attribute any gain to behavioural analytics specifically — and that is your entire claim**

7.9 Experiments needed to prove behavioural analytics improves adaptation

I Four studies, in order. Studies 1 and 2 are offline and de-risk everything downstream; do not build the adaptation engine before they pass.

1. **Study 1 — Offline predictive value (de-risking).** On public data with timing (EdNet, ASSISTments) plus your own instrumented pilot, run the ablation ladder under the leakage-safe pyKT protocol. *Success criterion: the behavioural increment must be statistically significant with a confidence interval that excludes the ~1–2% range attributable to protocol variation.* H Expect a modest effect; SAINT+'s 1.25% is the honest reference point.
2. **Study 2 — State validity.** Collect in-the-moment confidence probes on a random 10–15% of items, periodic effort/fatigue probes, and RTE-based engagement indices. Report correlation and calibration of each state head against its label, and apply Wise & Kong's criteria to the engagement estimator — including the negative criterion that it should *not* correlate with ability. *This study alone answers Gap 2 and is publishable independently.*
3. **Study 3 — Counterfactual policy evaluation.** Log data under a randomised or epsilon-greedy policy, then compare candidate policies offline using importance-weighted or doubly-robust estimation. I This is where hint-request endogeneity bites — plain replay will mislead you.
4. **Study 4 — Randomised controlled trial.** Randomise learners between the correctness+RT policy and the full behavioural-state policy, with identical content and identical policy architecture. Primary outcome: post-test learning gain. Secondary: items-to-mastery, session completion, dropout, and subgroup effects by prior ability. I M12 found an RL tutor helped lower performers specifically — **power your analysis for subgroups, because an aggregate null can conceal a real effect.** Follow M13's design as your template.

Realistic expectation, stated plainly

H Based on the effect sizes actually reported in this literature — SAINT+ at +1.25% AUC V, ITS meta-analytic effects around g ≈ .32–.37 S, contextual bandits at +15.2% skill gain over non-contextual baselines S — **the most likely outcome of your Study 4 is a small positive effect concentrated in a subgroup, most plausibly low-prior-ability or low-engagement learners.** Design and power for that outcome rather than for a large main effect. A well-powered, honestly-reported small effect with a validated state model is a substantially better paper than a large effect from an underpowered study with unvalidated constructs.

Would This Actually Be Novel?

The honest answer

As proposed: no. Substantially reformulated: yes, moderately.

Your proposal, read literally, is: collect performance and interaction signals, infer knowledge, confidence, engagement, fatigue, cognitive load and uncertainty, and use them to decide the next question, difficulty change, hint, prerequisite, repetition or break. Almost every element of that sentence has been done.

The three papers that most threaten your novelty claim

1. DASKT (K18) — IEEE TKDE, 2025 V

What it does that you proposed to do: extracts affective factors from non-affect-oriented behavioural data, simulates dynamic affective states (frustration, concentration, boredom, confusion) as the learner works, fuses them into knowledge state estimation, and reports beating state-of-the-art KT while retaining interpretability.

What is different in yours: DASKT stops at the knowledge state and prediction. It does not select actions, does not validate the affective states against labels (it simulates them, by its own framing, because affect labels are unaffordable), and does not use motor-level behaviour. I **Is the difference enough?** Yes, but only along the action and validation axes. If your contribution is "we fuse inferred affective/behavioural states into the learner model", DASKT has it and is in a top-tier venue.

2. LPKT (K9) and SAINT+ (K8) — KDD 2021 and LAK 2021 S V

What they do: make behavioural timing architectural rather than auxiliary. LPKT's core unit is literally *(exercise, answer time, answer)* with learning and forgetting gates; SAINT+ fuses elapsed and lag time into response embeddings.

What is different in yours: the *set* of behavioural signals (motor-level, not just timing) and the downstream use. I **Is the difference enough?** On its own, no. The architectural idea "behaviour belongs inside the state transition" is taken.

3. The nonlinear fatigue state-space model (B32) — *Computers*, 2025 S**, unverified**

What it does: models fatigue as a latent state with load–recovery dynamics coupled to attention and learning, and derives interventions that regulate load, pacing and recovery scheduling.

What is different in yours: unknown until you read it. I **This is the biggest open risk in your entire proposal.** If that paper validated on real learners with an intervention study, your fatigue contribution collapses to a replication. If it is simulation-only, your empirical validation becomes a genuine contribution. **Read it before writing a single line of your proposal.**

Classification of the available contribution

Type	Applies?	Reasoning
Novel algorithm	No	A shared encoder with multiple supervised heads feeding a contextual bandit is standard machine learning composition. Nothing in the mathematics is new, and claiming otherwise invites a reviewer to name the components
Novel feature combination	Yes — qualified	Cursor trajectory features combined with timing, attempts and answer-change sequences *inside a knowledge-tracing and item-selection pipeline* is a combination I found no instance of. But each

Type	Applies?	Reasoning
		component exists separately, and the combination is novel only because nobody has tested whether it helps — not because it is conceptually difficult
Novel learner model	**Partially**	Multi-state estimation exists in pieces. What is genuinely absent is a model whose states are *separately validated against independently collected labels*. The novelty is in the validation regime, not the state set
Novel system architecture	**Weakly**	The pipeline is the standard sense–model–decide loop. Architecture diagrams are not contributions. I What can be a contribution is the *state-interaction policy rules* — low confidence with high engagement warrants a hint, low confidence with low engagement warrants difficulty reduction, productive confusion warrants leaving the learner alone (grounded in B9, B10, B5)
Novel application	**Yes**	Transferring validated mouse-tracking uncertainty methodology from cognitive psychology and survey methodology into adaptive learning is a legitimate cross-domain application contribution
Incremental improvement	**Yes — this is the honest primary label**	On the prediction axis, your work will be incremental relative to SAINT+, LPKT and DASKT. Accept this rather than fight it
No meaningful novelty yet	**Applies to the proposal as literally worded**	"Use behavioural analytics to improve adaptive learning" is a description of an existing research area, not a research gap

The defensible novelty statement

Do not write: "We propose a novel adaptive learning system using deep behavioural analytics." Write something closer to:

> **"Behavioural signals are widely used to improve next-answer prediction in knowledge tracing, but it has not been shown that they improve instructional decisions, and the latent states they are said to represent are rarely validated against independent measurements. We instrument an adaptive learning platform to collect confidence, effort and difficulty labels alongside fine-grained interaction telemetry including cursor trajectories; we validate four learner-state estimators against those labels; and we test, in a randomised trial with the policy architecture held constant, whether conditioning adaptation on the validated behavioural state improves learning gain and instructional efficiency over correctness and response time alone."**

I That claim is narrower, fully supported by the gaps evidenced in Part 5, and it cannot be defeated by pointing at SAINT+, LPKT or DASKT — because none of them did any of it. **It also has the property that a null result is still a publishable finding**, which is the best structural protection a thesis can have.

Final Deliverables

D1. Top 10 papers to read personally

#	Paper	Why it is important	Relates to	What to pay attention to
1	**DASKT** — arXiv:2502.10396 / TKDE V	Closest published relative of your whole proposal, in a top venue	Your entire premise; Gaps 2 and 5	Exactly how affect is extracted without affect labels; whether affect states are validated or only simulated; whether anything downstream of prediction is done. **Your differentiation paragraph will be written against this paper**
2	**pyKT** — arXiv:2206.11460 V	Defines what a credible KT evaluation now looks like	All of Part 7's methodology	The specific leakage mechanism and the evaluation protocol. **Adopt the protocol and say so explicitly in your methods section**
3	**Gervet et al., When Is Deep Learning the Best Approach to KT?** — *JEDM* 12(3):31–54 ERIC / PDF V	Tells you when your deep model is justified at all	Model choice (7.7); baselines (7.8)	The dataset-property conditions favouring LR versus DKT, the ablations, and the calibration analysis. **Check which condition your dataset falls into before choosing an architecture**
4	**SAINT+** — arXiv:2010.12042 V	The honest magnitude of a pure timing-feature gain: +1.25% AUC	Gap 7; your effect-size expectations	How elapsed and lag time are embedded rather than concatenated. **Calibrate your expectations to 1.25%, not to DKT's disputed 25%**
5	**LPKT** — KDD 2021 S	Behaviour as architecture, not as auxiliary features	Gaps 4, 5, 7	The learning gate / forgetting gate construction over (exercise, answer time, answer). **The nearest architectural precedent to what you intend to build**
6	**Fatigue state-space model** — *Computers* 15(6):350 S **unverified**	Decides whether your fatigue contribution survives	Gap 4	**One question above all: real learners or simulation?** Then: how fatigue is identified, and whether the interventions were actually deployed and measured
7	**Wise & Kong response-time effort**, via Large-scale Assessments in Education and Wise 2017 S	The only rigorous validation protocol for a behavioural construct in this field	Gap 2; Study 2	The five validity criteria — especially that the effort index should *not* correlate with ability. **Apply these to your engagement estimator verbatim; it is the cheapest rigour available to you**
8	**Sensor-free affect detection SLR** — arXiv:2310.13711 V (preprint)	The mature literature your affect claims will be judged against	Gaps 2, 5; Category E	How labels are obtained, which states are reliably detectable, and the stated need for shared log-plus-label databases. **Check whether a peer-reviewed version now exists**
9	**Belfer, Kochmar & Serban, contextual bandits** — arXiv:2207.14003, AIED 2022 V	A deployed adaptation policy with a real RCT	Gap 1; Study 4; policy choice	The RCT design, the reward definition, and how the bandit context was constructed. **Your evaluation should look like theirs with a richer context vector**
10	**Mouse tracking as a window into decision making** — Behavior Research Methods, with the reproducibility critique S	The methodological foundation of your most novel signal	Gap 3; Study 1	Which trajectory measures are validated versus ad-hoc, and the reproducibility warnings. **Pre-register your feature definitions from this before you collect data**

Two honourable mentions worth your time: the telemetry-versus-self-report negative result (arXiv:2606.28881, preprint) because it is the strongest published argument *against* your confidence module and you must answer it; and Hakimi, Eynon & Murphy on digital-trace ethics because a system built on keystroke and cursor capture needs a serious ethics section, not a paragraph.

D2. Top 3 research gaps, ranked

1. **Gap 1 — The estimation–action gap.** Strongest. Real, acknowledged in recent literature, feasible at your scale, and publishable whichever way the result falls.
2. **Gap 2 — Latent states asserted without ground truth.** Nearly as strong, and it is what makes Gap 1's answer credible. Requires that you commit to label collection *from the first day of data collection* — it cannot be retrofitted.

3. **Gap 3 — Micro-interaction signals absent from adaptive pipelines.** Sharpest single empirical question, highest upside, and the most likely to return a modest or null answer. Scope it to desktop interaction explicitly.

D3. Proposed research framework

Your original pipeline is sound in outline but linear, and it omits the three things that determine whether the work is publishable: labels, calibration, and offline policy evaluation. Revised:

```
                    ┌──────────────────────────────────────────┐
                    │  LEARNER INTERACTION (web ALP, desktop)  │
                    └───────────────────┬──────────────────────┘
                                        │
         ┌──────────────────────────────┼──────────────────────────────┐
         │                              │                              │
         ▼                              ▼                              ▼
┌──────────────────┐      ┌────────────────────┐     ┌──────────────────────┐
│BEHAVIOURAL DATA  │      │ LABEL ACQUISITION  │     │ CONTENT / ITEM BANK  │
│ correctness      │      │ confidence probe   │     │ Rasch-calibrated     │
│ response time    │      │  (10–15% items)    │     │ difficulty, KC map   │
│ decision latency│      │ effort probe       │     └──────────┬───────────┘
│ attempts, hints  │      │ difficulty rating  │                │
│ option changes   │      │ (optional: BROMP   │                │
│ cursor traject.  │      │  observation)      │                │
│ idle + focus     │      └─────────┬──────────┘                │
└────────┬─────────┘                │                           │
         │                          │  supervises               │
         ▼                          │                           │
┌────────────────────────────┐      │                           │
│FEATURE / REPRESENTATION    │      │                           │
│per-item normalisation,     │      │                           │
│pre-registered trajectory   │      │                           │
│features, sequence encoder  │      │                           │
└────────┬───────────────────┘      │                           │
         │                          │                           │
         ▼                          ▼                           │
┌───────────────────────────────────────────────────────┐       │
│ LEARNER STATE ESTIMATION — separately supervised,     │       │
│ separately CALIBRATED heads, each with uncertainty    │       │
│                                                       │       │
│  knowledge      engagement     confidence   fatigue   │       │
│  (correctness)  (RTE-valid.)   (probes)     (decay)   │       │
└────────┬──────────────────────────────────────────────┘       │
         │                                                      │
         │  ┌─ VALIDATION GATE ─────────────────────────────┐   │
         │  │ A state may enter the policy ONLY if it passes│   │
         │  │ its validity check against labels. Unvalidated│   │
         │  │ states are reported, not acted upon.          │   │
         │  └───────────────────────────────────────────────┘   │
         ▼                                                      │
┌───────────────────────────────────────────────────────┐       │
│ ADAPTIVE DECISION ENGINE (contextual bandit)          │◄──────┘
│ state-interaction rules, e.g.                         │
│  low conf + high engage      → hint                   │
│  low conf + low engage       → reduce difficulty      │
│  confusion + rising accuracy → DO NOT intervene       │
│  high attempts + flat trend  → prerequisite           │
│  within-session decay        → suggest break          │
│ constrained action space + safety floor               │
└────────┬──────────────────────────────────────────────┘
         │
         ├──────────────► ┌────────────────────────────┐
         │                │ EXPLANATION: decision      │
         │                │ decomposed over named,     │
         │                │ validated state deltas     │
         ▼                └────────────────────────────┘
┌──────────────────────┐         ┌──────────────────────────────────┐
│ NEXT LEARNING ACTION │────────►│ LOGGING for OFFLINE POLICY EVAL  │
```

```
 └───────────────────┘         │ (propensities recorded; IPS/DR) │
          │                    └─────────────────────────────────┘
          └────────── feedback loop ──────────▶ (back to interaction)
```

Four changes from your original diagram, and why each matters: (1) **label acquisition is a first-class input**, not an afterthought — this is Gap 2 and it must be designed in from the start; (2) **a validation gate** sits between estimation and action, so unvalidated states are reported but never acted on — this is what makes the system defensible; (3) **propensities are logged** so offline policy evaluation is possible, which you cannot add retroactively; (4) **explanation branches off the decision**, not off the prediction, because that is where the gap actually is.

D4. Recommended research methodology

Element	Recommendation
Dataset strategy	**Two-tier.** Tier 1: public datasets for the KT-model component — EdNet for timing at scale (K25), ASSISTments for attempts and hints. These give comparability with published baselines. Tier 2: your own instrumented ALP for signals no public dataset contains — cursor trajectories, option-change sequences, confidence probes. `I` **You cannot answer Gaps 2 and 3 from public data; no public dataset has the labels or the motor telemetry.** Budget accordingly: your own data collection is the critical path, not the modelling
Sample	Power the RCT for the realistic effect size — around g ≈ .3 from the ITS meta-analyses (M19) — not for a hoped-for large effect. Plan subgroup analysis by prior ability from the start (M12). Report the power calculation
Experimental design	Four studies in the order given in 7.9: offline ablation → state validity → offline policy evaluation → RCT. **Each stage gates the next.** If Study 1 shows no incremental value from trajectory features, that is a finding, and you reallocate to Gaps 1 and 2 rather than proceeding on a dead signal
Baselines	As in 7.8. The two non-negotiable ones are **feature-engineered logistic KT** and **SAINT+ or LPKT**. Omitting either is fatal for a different reason: the first because it may simply beat you, the second because it already has behavioural features
Evaluation metrics	*Prediction:* AUC, accuracy, RMSE, plus **calibration (ECE / reliability curves)** — calibration matters more than AUC when the output drives a decision. *State validity:* correlation and calibration against probes; Wise & Kong's criteria for engagement. *Decision quality:* **learning gain (pre/post), items-to-mastery, time-to-mastery, session completion, dropout.** *Fairness:* performance parity across prior-ability groups and, where available, demographic groups — B25 shows input dynamics can encode demographics
Statistical validation	Mixed-effects models with random intercepts for learner and item — **interactions are nested within learners and treating them as independent inflates significance**, a common error in this literature. Report effect sizes with confidence intervals, not p-values alone. Correct for multiple comparisons across the ablation ladder (Holm). **Pre-register hypotheses, trajectory feature definitions and the analysis plan before collection** — this directly answers the reproducibility critique in B19 and costs you nothing
Ablation studies	(a) Signal-class ladder: correctness → +RT → +attempts/changes → +trajectory → +typing (if applicable). (b) State-head ablation: which states, once removed, degrade *decision* quality — not just prediction. (c) Policy ablation: same state, different policies, to separate state contribution from policy contribution. `I` **Ablation (b) is the one that produces your headline result; most papers only run (a)**
Ethics and privacy	Explicit informed consent following the model in M23; report the privacy–utility curve from Gap 8; store aggregated trajectory features rather than raw cursor streams where possible; justify each collected signal by its measured contribution. `I` **If a signal does not earn its place in the ablation, remove it from the system — that is both better ethics and a stronger paper**
Threats to validity to state	Desktop-only motor signals; single-domain content; probe reactivity (asking about confidence may change it); Hawthorne effects; hint-request endogeneity under an adaptive policy; EdNet's TOEIC-specific, single-country population

D5. Reference list

Grouped as in Part 3. Every URL below was returned by a search or retrieved during this review. **Where the tag is `S`, you must open the source and confirm the details before citing it in a submitted paper.** Items marked "preprint" are not peer-reviewed; check for a published version. Where authorship could not be established from retrieved material, it is left blank rather than guessed.

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Limitations of this review

Stated plainly, because a review that hides its own limitations is exactly the kind of document this review criticises.

- **This is not a PRISMA systematic review.** It is a structured, evidence-tagged critical review. There was no pre-registered protocol, no formal inclusion/exclusion screening log, and no dual-reviewer agreement statistics. If your paper requires a formal systematic review, this document is the input to one, not the output of one.
- **Search was English-language and web-index-based.** Paywalled full texts were largely inaccessible; several publishers (MDPI, ACM DL, parts of Elsevier) actively blocked retrieval, which is why some of the most relevant items carry S tags rather than V.
- **Most claims are abstract-level.** Where a finding rests on a full-text detail I could not retrieve, the document says so rather than filling the gap.
- **Citation counts and some author lists are omitted deliberately.** They were not retrievable, and an estimated citation count or a guessed author list is a fabrication, which would defeat the purpose of the exercise.
- **Recency asymmetry.** Several highly relevant 2025–2026 items are preprints and are labelled as such; their claims may not survive peer review.

Before submitting anything based on this document: retrieve and read the ten papers in D1, confirm every S-tagged claim you intend to cite, and check whether the preprints have peer-reviewed versions. Treat this review as a map of the territory and a list of things to verify — not as a citable source in itself.